

ChainFL-RA: Blockchain-Enabled Decentralized Federated Learning With Reputation-Weighted Robust Aggregation and On-Chain Auditable Incentives

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ABSTRACT Federated learning (FL) concentrates three risks in one place: the aggregator is a single point of trust, Byzantine clients can poison the global model, and honest contributors lack auditable credit. Existing defenses address these risks in isolation: robust aggregation ignores accountability, blockchain logs record without influencing learning, and incentive layers rest on unverifiable signals. ChainFL-RA unifies them under one principle—a single verifiable quality signal drives the entire trust loop. The same per-update evidence (cosine similarity to a momentum-anchored median reference, guarded by a magnitude check) determines aggregation weights, updates a slash-based on-chain reputation, and settles token rewards and penalties through a smart contract, with every decision committed to a hash-chained ledger. The rule requires neither server-side validation data nor knowledge of the adversary count. Evaluated against FedAvg, Krum, FedRoLA, and FLAME under label-flipping and update-scaling at 10–30% malicious ratios (full matrix on MNIST and CIFAR-10, 30% cells on Fashion-MNIST, 3–5 seeds) and an adaptive Min-Max attack, ChainFL-RA is tied with the best baseline without attack and leads 15 of 19 comparison cells, including all seven CIFAR-10 cells: it retains 96.94%–97.36% under scaling where FedAvg collapses below 9.7%, and attains 96.78%/93.74% under Min-Max at 20%/30% adversaries. Slash precision exceeds 0.92 with honest false-slash rates below 0.9%. A gas model places the audit layer at 0.06M–1.1M gas and about 2.2 s latency per round, indicating that fully auditable decentralized FL is practical at cross-silo scale.

INDEX TERMS Federated learning, blockchain, robust aggregation, reputation mechanism, poisoning attack, incentive mechanism.

I. INTRODUCTION

1) A. Background and Motivation

Federated learning (FL) lets data owners jointly train a model without sharing raw data: clients keep their data locally and share only model updates with an aggregation server [1], [2]. The convenience has a price: three systemic risks concentrate in one logical aggregator. Should that aggregator be compromised, go down, or simply grow lazy, the whole federation stalls or drifts — and nothing in the protocol notices [3]. A second exposure comes from the clients themselves; a minority of Byzantine participants submitting crafted updates can bend the global model toward their own objectives [4], [5]. The third risk is quieter: honest contributors accumulate

no verifiable credit, dishonest ones risk no credible penalty, and the aggregation process stays opaque to outsiders [6], [7].

Every one of these risks has a familiar countermeasure; every countermeasure, in turn, has a blind spot. Robust estimators and score-based filters address poisoning, but they break precisely where realistic deployments live: under non-IID data, honest updates become mutually dissimilar, Krum-style selection discards most honest information per round (losing up to 13.4 points of accuracy under label skew in our measurements), and validation-based defenses such as FLTrust [8] assume a clean server-side root dataset that a decentralized consortium does not possess. Blockchain-based systems write model hashes to a ledger [3], [9], but the ledger

is a passive log that records without influencing learning, so a detected poisoner keeps participating. Reputation and contract-based mechanisms [6], [7] reward participation, yet their reputation signals usually require trusted validation data and their settlement is disconnected from the aggregation rule. Strip away the surface differences and a single root cause remains: *the quality signal, the aggregation decision, the accountability record, and the incentive settlement are four separate subsystems with four separate trust assumptions.*

Blockchain offers a substrate for unifying them: a replicated, append-only ledger with smart-contract programmability can replace the single aggregator with a decentralized, tamper-evident coordination layer [3], [9]. A growing body of work explores this synergy (blockchain-assisted reputation management [10], poisoning defense with on-chain incentives [11], [12], verifiable secure aggregation [13], [14]), but three gaps remain open. (G1) **Robustness without a trusted reference**: most robust rules need the adversary count, need clean validation data, or collapse under non-IID heterogeneity; direction-based evidence such as cosine similarity to a robust reference needs no held-out data [15], but a rule that exploits it while needing no adversary count and tolerating label skew is missing. (G2) **A closed loop rather than a log**: no existing system lets the *same* measured quality signal drive aggregation weights, reputation, and token settlement, so accountability never becomes causal. (G3) **Quantified practicality**: the on-chain cost of such loops (gas, storage, latency per training round) is rarely quantified end-to-end, leaving deployability uncertain.

2) B. Contributions

ChainFL-RA, the framework we develop here, is our attempt to close all three gaps under that single unifying principle. Its contributions line up with the gaps one-to-one:

1. **Reputation-weighted robust aggregation (RAgg), which closes G1.** At the heart of RAgg lies a coordinate-wise median reference anchored to an exponential moving average of previously accepted aggregates. Every update passes two complementary gates — cosine similarity against that reference, and a magnitude check that is active from round one — before entering a slash-only reputation memory with slow recovery; tempered weights then produce the aggregate. Slashed clients lose influence multiplicatively, so the rule accumulates evidence of Byzantine behavior across rounds without amplifying transient signals; it requires no server-side validation data and no knowledge of the number of adversaries.
2. **An on-chain auditable incentive layer, closing G2.** Each round ends with the global model hash, per-client aggregation weights, quality scores, and reputation transitions written to a hash-chained ledger; a smart contract then settles token rewards and slashes using exactly the evidence that produced the aggregate — accountability becomes causal rather than forensic. A deterministic in-process simulator realizes the layer,

modeling block structure, hash chaining, gas accounting, and consensus latency faithfully enough that trustlessness costs stay measurable and reproducible.

3. **Empirical evaluation, addressing G3.** We implement four baselines spanning the recent robust-aggregation landscape (FedAvg [2], Krum [5], FedRoLA [22], and FLAME [54]) and evaluate all methods on MNIST, Fashion-MNIST, and CIFAR-10 under label-flipping, update-scaling, and an adaptive Min-Max attack at 10%–30% malicious ratios over five seeds, with module-level ablations, detection-level metrics, adversarial-weight dynamics, and per-round gas, storage, and latency quantification. All experiments are fully scripted and reproducible from the released artifacts.

3) C. Application Scenarios

The design point of ChainFL-RA is the cross-silo federation: a small to medium number of institutional participants (hospitals, banks, regional grid operators, or IoT platform providers) that hold substantial, statically partitioned datasets and can run infrastructure-grade nodes. Three concrete scenarios motivate the trust loop. In multi-hospital imaging consortia, a hospital that silently trains on mislabeled scans must be detectable and penalizable, because diagnostic quality is a shared liability; an on-chain reputation trail gives regulators and consortium partners an auditable basis for membership decisions. In inter-bank fraud detection, participants are simultaneously collaborators and competitors, so the incentive layer must make honest contribution economically rational and sabotage economically irrational; tokenized settlement on a ledger that no single competitor controls is the natural mechanism. In IoT edge intelligence, gateways aggregate models from heterogeneous device fleets, and the ledger doubles as a provenance record that downstream consumers can consult before trusting a model [18], [19]. Section VI-G maps the experimental findings back to these three scenarios.

4) D. Organization

Section II reviews related work. Section III formalizes the system and threat models. Section IV presents the ChainFL-RA design, including the RAgg rule, its convergence analysis, the on-chain layer, a complexity analysis, and the composition with privacy mechanisms. Section V describes the experimental setup. Section VI reports and analyzes results, including ablations, overhead, and application implications. Section VII discusses limitations and future directions, and Section VIII concludes.

II. RELATED WORK

1) A. Federated Learning and Its Robustness Problem

Since the introduction of FedAvg [2], FL research has progressed along optimization, systems, and security axes. FedProx adds a proximal term to stabilize local training under heterogeneity [16]; SCAFFOLD introduces control variates to correct client drift [17]; and the non-IID challenge has been

surveyed extensively [4], [1], establishing that statistical heterogeneity (not partial participation or communication constraints) is the dominant source of accuracy loss in realistic federations. On the security axis, Byzantine-robust aggregation replaces the mean with robust estimators; coordinate-wise medians, geometric medians, and nearest-neighbor scoring rules such as Krum are representative, and large-scale empirical studies have mapped their strengths and failure modes [5], [20]. A consistent finding of these studies is that no single estimator dominates: rules that excel under IID data can underperform the plain mean under label skew, because honest updates themselves become mutually dissimilar. Cosine-similarity-based filtering has recently emerged as an effective signal against both data and model poisoning [15], precisely because direction agreement is comparatively stable across honest-but-skewed clients, and contribution-aware variants address class imbalance by weighting updates according to estimated class coverage [21]. Neighboring defenses share this directional-evidence view: RFA aggregates through a smoothed geometric median [53], FLAME combines clustering with norm clipping [54], and FLOW pairs per-round cosine detection with cross-round punishment rather than removal [55], the closest algorithmic neighbor of our cosine-plus-reputation combination. The strongest attacks against such rules are themselves geometry-aware [50], [51], which motivates our dual-channel (direction and magnitude) design, and history-based defenses such as Byzantine momentum [52] show that cross-round memory can stabilize robust estimation, the property our anchored reference exploits. Layer-based robust aggregation against model poisoning [22], dual-refined aggregation that looks beyond update divergence [23], and dynamic trust-adaptation defenses for asynchronous federations [24] further confirm that the granularity at which robustness is enforced matters. Robustness has also been studied for specific application verticals, including network intrusion detection [25], [26], and edge-deployed FL protocols where hierarchy and asynchrony complicate aggregation [18], [19], [27], [28]. Our work differs in that robustness signals are not only used for aggregation but are also written to a ledger that drives incentives, closing the accountability loop: the same cosine-quality evidence that steers the aggregate also steers reward and slashing, so that robustness has an economic memory.

2) B. Blockchain-Enabled Federated Learning

Integrating blockchain with FL has been surveyed in depth [3], which catalogs architectures by where the ledger sits in the training loop (as a model store, as a coordination fabric, or as an incentive rail) and identifies open problems in scalability, privacy, and governance. Early systems established the feasibility of on-chain model exchange and incentive settlement; DeepChain introduced auditable, privacy-preserving training with blockchain-based incentives and remains the reference point for accountability-by-design [9], blockchain-enabled frameworks have since coupled decentralization with incentive settlement and peer-consistency checks [29], and

on-chain poisoning defense with gradient-inversion resistance has been shown [30]. Biscotti established peer-to-peer blockchain FL with privacy protection and evaluated 30% adversaries [57]; FLEDGE couples ledger-native cryptocurrency rewards and penalties with poisoning mitigation on MNIST, Fashion-MNIST, and CIFAR-10 [56]; and PBFL combines blockchain coordination with cosine-based malicious-gradient judgment and homomorphic secure aggregation [58]. Recent work tightens the coupling between ledger and learning. T-BFL maintains decentralized reputation that steers both aggregation and consensus difficulty, so that learning quality and block production share one trust currency [10]; BPFL detects poisoned gradients via cosine similarity under Paillier encryption and penalizes adversaries through deposits, showing that poisoning defense and incentive settlement can share one evidence stream [11]; Dong et al. combine peer voting with reward-and-slash smart contracts, proving convergence guarantees under a stake-weighted adversary model [12]; and BSR-FL achieves Byzantine robustness with functional encryption over a blockchain-coordinated protocol, showing that privacy and robustness can be co-designed rather than bolted together [31]. Robust blockchain-based federated architectures have also been prototyped for decentralized split learning and for industrial consortium settings. Reputation-aware revenue allocation and monetization further illustrate the value of explicit trust accounting [32], [33], [34], hierarchical variants extend the idea to multi-tier edge topologies [32], and application-driven integrations such as blockchain-assisted medical classification illustrate the cross-domain demand for auditable federations [35]. ChainFL-RA shares this agenda but differs in three respects: the reputation signal is derived from robust geometry rather than votes or probing accuracy; the same signal drives aggregation weights, reputation transitions, and token settlement in one closed loop; and the ledger cost is reported per round with a reproducible gas model rather than asserted qualitatively.

3) C. Incentives, Verifiability, and Privacy Mechanisms

Incentive design for FL spans contract theory, auctions, and reputation. Truthful and reliable mechanisms based on reputation and contract theory motivate sustained participation [7], [6], [36], [37], [38], and blockchain-native trust-management designs make interactions and settlement transparent [39]. Verifiable aggregation has advanced rapidly: VeSAFL and LVSA provide verifiable secure aggregation with different efficiency trade-offs [13], [40]; malicious-resistant non-interactive verification [14], publicly verifiable aggregation [41], transcript-bound verification with differential privacy [42], and traceable edge-assisted aggregation [43] extend verifiability along complementary dimensions. Smart-contract-assisted secure aggregation has also been reported [44]. Privacy mechanisms (differential privacy and secure aggregation among them) are largely orthogonal to our design and can be composed with it [45]; gradient-purification defenses for decentralized FL address related threats in serverless topologies [46], and fake-client untargeted attacks motivate

defenses that, like ours, accumulate evidence across rounds [47]. Homomorphic-encryption-based protection with reputation has been combined with blockchain incentives as well [48]. Proof-of-reasoning consensus hints at tighter couplings between learning and consensus [49]. Our work is complementary: we focus on the quality–reputation–settlement loop and its robustness implications, and we deliberately keep the cryptographic layer simulated so that the systems loop can be studied in isolation.

III. SYSTEM MODEL AND PROBLEM FORMULATION

1) A. System Model

We consider a cross-silo federation of N clients $\mathcal{C} = \{1, \dots, N\}$, each holding a private dataset $\mathcal{D}_i$ of size $n_i = |\mathcal{D}_i|$ drawn from a distribution $\mathcal{P}_i$ over the sample space $\mathcal{X} \times \mathcal{Y}$. Distributions may be heterogeneous, $\mathcal{P}_i \neq \mathcal{P}_j$, which we instantiate with Dirichlet label-skew partitions in Section V. No central aggregator exists. Clients instead interact through a blockchain with smart-contract support (an Ethereum-style account-and-gas platform in the cost model of Section IV-D; validator operation may be drawn from the participants themselves or from a public validator set); the ledger provides (i) a total order of training rounds, (ii) tamper-evident storage of hashes and scores, and (iii) deterministic smart-contract execution for settlement. Learning proceeds in synchronous rounds $t = 1, \dots, T$. In each round, all clients train from the current global model w^t on their local data for E epochs, producing updates $\Delta_i^t = w_i^t - w^t$; updates are aggregated into a new global model, and the round's audit record is committed on-chain.

Aggregation execution model. Because only hashes and scalar records are stored on-chain, the physical aggregation locus must be made explicit. In ChainFL-RA, clients submit their updates to the validator committee over authenticated off-chain channels, together with an on-chain commitment $\text{SHA256}(\Delta_i^t)$. Every validator then independently executes RAgg over the same update set (whose integrity is pinned by the commitments), and the resulting aggregate is accepted only when an honest majority of validators obtains the identical output hash. Deterministic re-execution replaces trust in any single compute node, which is precisely how the single point of trust is removed without moving model payloads on-chain. A validator that deviates (e.g., tampers with scores) produces a mismatching hash and is excluded from the round by consensus. This model adds a factor- V replication of the (cheap, $O(Nd)$) aggregation computation across V validators and requires off-chain update distribution, whose bandwidth we account for in Section VI-F.

The design space rests on two trust assumptions. The first concerns the ledger: its consensus is honest-majority, validators may be curious but do not rewrite history, so anything committed is immutable and anything correctly computed by the contract is faithfully executed. This is the standard assumption of consortium chains and isolates our problem (the trustworthiness of *learning contributions*) from the orthogonal problem of consensus security. Second, clients are

economically rational: they respond to rewards and penalties, but a bounded fraction may be Byzantine in the sense of deviating arbitrarily from the protocol, whether from compromise, misconfiguration, or malice. Rational-but-honest clients fall to the incentive layer, Byzantine ones to the robust aggregation layer — and because both layers read the same quality signal, no client can be simultaneously learning-attackable and economically invisible.

The FL objective is the standard empirical risk minimization over the federation:

$$\min_{w \in \mathbb{R}^d} F(w) = \sum_{i=1}^N \frac{n_i}{\sum_j n_j} F_i(w), \quad F_i(w) = \frac{1}{n_i} \sum_{\xi \in \mathcal{D}_i} \ell(w; \xi), \quad (1)$$

where ℓ is the per-sample loss. In the presence of adversaries, honest aggregation approximates the mean over the honest subset $\mathcal{H}$ only; the design goal is a rule whose output remains close to $\sum_{i \in \mathcal{H}} \frac{n_i}{n_{\mathcal{H}}} \Delta_i^t$ under the attack model below.

2) B. Threat Model

A fraction $f \in [0, 0.3]$ of clients is Byzantine. Byzantine clients may (i) **label-flip**: train on locally relabeled data $y \mapsto (K - 1) - y$ for K classes, implanting a systematic error; or (ii) **update-scale**: train honestly but submit $\tilde{\Delta}_i = s \Delta_i$ with a large negative scaling factor s , directly steering the aggregate. Adversaries know the aggregation rule and may adapt within these two families; they do not collude beyond sharing the attack type, and they cannot forge ledger entries. Section VI-B additionally evaluates a Min-Max adaptive attack under a *stronger* adversary that observes all honest updates of the round, a stress test deliberately placed outside the threat model above and stated explicitly there. The aggregator function must satisfy: (a) **no validation data**, i.e., no clean held-out set is available to score updates; (b) **unknown f** , i.e., the exact number of adversaries is not assumed; and (c) **accountability**, i.e., every scoring and settlement decision must be reconstructible from the ledger.

3) C. Design Goals

ChainFL-RA targets four design goals: **(D1) robustness**, i.e., test accuracy under attack remains close to the attack-free level at $f \leq 0.3$; **(D2) heterogeneity tolerance**, i.e., the rule does not punish honest but statistically divergent updates under non-IID partitions; **(D3) auditability and incentives**, quality scores, weights, reputation, and token flows are committed on-chain and settled automatically; **(D4) practicality**, per-round on-chain overhead (gas, bytes, latency) remains modest as N grows. (We label design goals D1–D4 to distinguish them from the research gaps G1–G3 of Section I.)

Formally, let $\mathcal{B} \subset \mathcal{C}$ denote the Byzantine subset with $|\mathcal{B}| = fN$, and let $\Delta_{\mathcal{H}}^t = \sum_{i \in \mathcal{H}} \frac{n_i}{n_{\mathcal{H}}} \Delta_i^t$ be the honest aggregate that an idealized rule would produce, i.e., the honest restriction of the global objective in Eq. (1). We say an aggregation rule $\mathcal{A}$ is (δ, f) -resilient for this round if $\|\mathcal{A}(\{\Delta_i^t\}) - \Delta_{\mathcal{H}}^t\| \leq \delta \|\Delta_{\mathcal{H}}^t\|$ for any admissible attack at ratio f ; in this work the

property is evaluated statistically (in expectation and in high probability over seeds) rather than as a worst-case guarantee. RAgg pursues this property constructively: as adversarial reputation decays through slashing, the adversarial mass in the weight vector of Eq. (6), $\sum_{i \in \mathcal{B}} \omega_i^t$, trends downward over rounds, with transient rebounds from the additive recovery path (we measure this quantity directly in Section VI-B (Fig. 4)) so the deviation above is driven toward zero even though no adversary is ever explicitly identified or excluded. This relaxation (soft suppression instead of hard filtering) is deliberate: hard exclusion requires a detection decision that can be wrong about honest outliers, whereas soft suppression bounds the damage of a misclassification to a partial down-weighting that later rounds can correct.

IV. THE CHAINFL-RA FRAMEWORK

1) A. Overview

ChainFL-RA interleaves an off-chain learning plane with an on-chain trust plane, as illustrated in Fig. 1. The system state at round t is the tuple $(w^t, \{r_i^t\}, m^t, B_{t-1})$: the global model, the reputation vector, the momentum anchor, and the ledger tip. One round of the unified loop maps this state to the next through four coupled transitions:

1. **Produce (learning plane).** Clients train from w^t and submit updates $\{\Delta_i^t\}$ with hash commitments.
2. **Evaluate (learning plane).** RAgg (Section IV-B, Eqs. (2)–(6)) computes the anchored reference from $\{\Delta_i^t\}$ and m^{t-1} , the quality vector $\{q_i^t\}$, the reputation update $\{r_i^t\}$, and the weights $\{\omega_i^t\}$, producing Δ^t and w^{t+1} .
3. **Commit (trust plane).** The block B_t seals $\text{SHA256}(w^{t+1})$ and the per-client records $\{(id_i, q_i, \omega_i, r_i)\}$; validators re-execute RAgg and accept the block only on majority agreement.
4. **Settle (trust plane).** The contract distributes rewards and slashes from the committed $\{q_i\}$, and the updated $\{r_i^t\}$ feeds back into step 2 of the next round.

The loop is closed by construction: the quality signal q_i is computed once and consumed three times, by the aggregator (as ω_i), by the reputation memory (as r_i), and by the settlement contract (as rewards and slashes), so no subsystem ever acts on evidence that another subsystem cannot see. Because the block commits to every quantity that determined the aggregate, any third party can replay the aggregation decision from the ledger and verify that settlement followed the published rule. The ledger is the audit trail of the learning process itself, echoing the auditability advocated in [9] and [10] but with geometry-derived rather than vote-derived evidence.

In the learning-to-ledger direction, every quantity that influenced the global model is committed, so the model's provenance is complete: given any deployed global model, one can enumerate the exact contributions that produced it, round by round, client by client. In the ledger-to-learning direction, the settlement outcome feeds back into participation decisions, and the slashed reputation feeds into the next round's weights, so the ledger is not a passive notary but an active state variable

of the learning dynamics. This bidirectional closure is what distinguishes ChainFL-RA from systems that record learning interactions on-chain for after-the-fact defense or forensics [30]: accountability here is causal and immediate, not retrospective.

2) B. Reputation-Weighted Robust Aggregation (RAgg)

Let $\{\Delta_i^t\}_{i=1}^N$ be the updates of round t , flattened to vectors in $\mathbb{R}^d$. RAgg proceeds in five steps.

Step 1, Anchored spectral reference. We compute the reference as a momentum-anchored average of K random-subsample geometric medians (RSGM). Drawing $K = 8$ subsets $S_1, \dots, S_K$, each of size $\lceil \rho(N + fN)/2 \rceil$ with $\rho = 0.75$ (every such subset contains an honest majority by the pigeon-hole principle whenever $f < 1/2$), we solve for each subset's geometric median $\tilde{\mu}_k$ by Weiszfeld iteration and anchor their average to an exponential moving average of past accepted aggregates:

$$\begin{aligned} \bar{\Delta}^t &= \lambda m^{t-1} + (1 - \lambda) \frac{1}{K} \sum_{k=1}^K \tilde{\mu}_k(\{\Delta_i^t\}_{i \in S_k}), \\ m^t &= \lambda m^{t-1} + (1 - \lambda) \Delta^t, \end{aligned} \quad (2)$$

with $\lambda = 0.6$ and m^0 initialized by the first round's reference. The geometric median has breakdown point $1/2$: within each subsample the honest majority pins $\tilde{\mu}_k$ inside the honest ball regardless of what the minority submits (Theorem 2), so the reference inherits an honest-majority guarantee that the plain coordinate-wise median of Eq. (2)'s predecessor only enjoyed per-coordinate; averaging over K independent subsets shrinks the reference variance by $1/K$ (Minsker-style concentration [59]). The momentum anchor contributes drift resistance against the transient-consistency failure mode we diagnosed empirically: when honest clients hold strongly skewed data, a *consistent* adversarial coalition can momentarily become the best-aligned cluster with respect to any instantaneous statistic, and the historical descent direction, accumulated from rounds in which the honest majority dominated, breaks this symmetry (the pure-RSGM ablation of Section VI-C quantifies exactly this failure). Using history to stabilize Byzantine-robust estimation parallels the worker-momentum analysis of [52], with the difference that our anchor mixes into the *reference* rather than the update.

Step 2, Cosine quality. Each update receives a clipped cosine-similarity quality score

$$q_i^t = \max\left(0, \frac{\langle \Delta_i^t, \bar{\Delta}^t \rangle}{\|\Delta_i^t\| \|\bar{\Delta}^t\| + \nu}\right) \in [0, 1], \quad (3)$$

where $\nu = 10^{-12}$ is a numerical stabilizer.

Directional (rather than magnitude) agreement is used because magnitude varies legitimately with local data size and heterogeneity, whereas direction is the carrier of poisoning intent; cosine signals have proven effective for poisoning detection in [15] and [11].

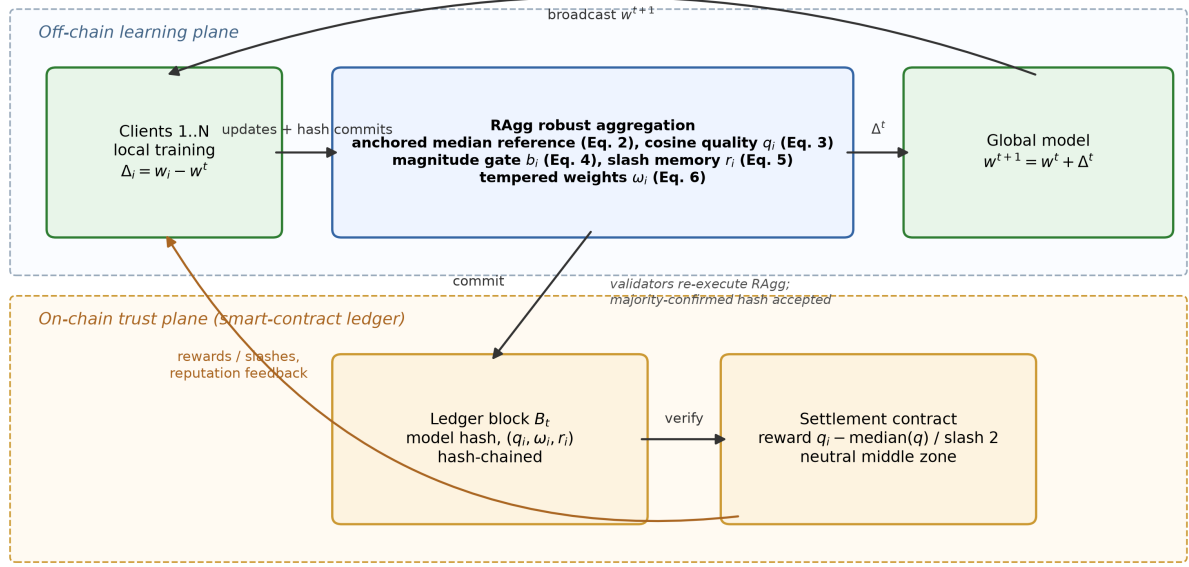


FIGURE 1. ChainFL-RA overview: an off-chain learning plane (local training, RAgg aggregation, model broadcast) closed through an on-chain trust plane (hash-chained audit blocks and contract-based settlement). One quality signal q_i drives weights, reputation, and settlement alike.

Directional evidence alone, however, has two blind spots that we diagnosed empirically (Sections VI-B and VII): near convergence, honest updates shrink and their directions become poorly conditioned, so cosine separability between honest and adversarial updates degrades; and during the warm-up rounds the direction gate is deliberately frozen. Update-scaling attacks exploit both windows, because their damage is carried by magnitude rather than by direction. We therefore complement the cosine score with a **smooth magnitude gate**

$$g_i^t = \sigma \left(s_0 \ln \frac{\kappa}{\rho_i^t} \right) \in (0, 1), \quad (4)$$

$$\rho_i^t = \frac{\|\Delta_i^t\|}{\text{median}(\{\|\Delta_j^t\|\}_{j=1}^N)},$$

with cap ratio $\kappa = 5.0$ and sharpness $s_0 = 3$: honest magnitudes ($\rho_i \approx 1$) pass with $g_i \geq 0.99$, norm ratios at the threshold receive $g_i = 0.5$, and a scaling attack inflating its norm tenfold passes with $g_i < 10^{-5}$, an *exponentially decaying* pass probability in the attack amplitude (Theorem 3). Unlike the hard indicator used in earlier versions, the smooth gate provides a nonzero-but-small pass rate for borderline magnitudes, which avoids a discontinuous trust cliff at κ while remaining effectively binary for the amplitude regimes of interest. The norm ratio is a within-round statistic that is meaningful as soon as the first updates arrive, so the gate needs no warm-up and closes the early-round window in which a scaling adversary could otherwise seize the momentum anchor before the direction gate becomes trustworthy.

Step 3, Slash-based reputation memory with slow recovery. Each client carries a reputation $r_i^t \in (0, 1]$, initialized

at $r_i^0 = 0.5$. Reputation can only decrease (multiplicatively, through slashing) and recovers additively at a slow rate:

$$r_i^t = \begin{cases} \gamma r_i^{t-1}, & q_i^t < \tau_t \vee g_i^t < 0.5 \quad (\text{slash}) \\ \min(1, r_i^{t-1} + \rho), & \text{otherwise (recovery)}, \end{cases} \quad (5)$$

$$\tau_t = \max(0.05, 0.25 \cdot \text{median}(\{q_j^t\})), \quad (6)$$

with slash factor $\gamma = 0.5$ and recovery rate $\rho = 0.05$. Because the absolute scale of the cosine scores of Eq. (3) drifts over training, the slash threshold is adaptive, tracking the round's median quality. The update rule deliberately avoids tracking the quality signal *upward* (e.g., by EMA): our pilot experiments showed that EMA-tracking creates a rich-get-richer feedback loop (transiently high-quality clients attract weight, the reference drifts toward them, their quality rises further) which concentrates the aggregate on few clients and can collapse training under non-IID data. Slash-only memory cannot amplify signals; it can only punish repeat offenders, so intermittent attackers accumulate penalties across rounds (addressing the fake-client dynamics noted in [47]) while honest clients, whose false-slash rate never exceeds 0.6% in any cell of our suite (Section VI-B), remain systematically undemoted (D2).

Step 4, Warm-up. In the first two rounds the cosine signal is not yet separable (freshly initialized models produce mutually dissimilar updates regardless of intent), so reputation is frozen and the directional channel is disabled; aggregation falls back to magnitude-gated sample weighting ($\omega_i \propto n_i g_i$ in Eq. (6)). The magnitude gate needs no warm-up (its within-round norm ratios are meaningful from the first updates),

so update-scaling adversaries are blocked from the warm-up window, an attack path we observed in pilot experiments, where ungated scaling updates poisoned the momentum anchor within two rounds and triggered a cascade of false slashes against honest clients.

Step 5, Dual-channel tempered aggregation. The quality gate is likewise smoothed: instead of the hard indicator $\mathbb{I}[q_i' \geq \tau_t]$, each update receives a directional pass rate $s_i^t = \sigma((q_i' - \tau_t)/T_t)$ with an *adaptive temperature* $T_t = \max(\text{IQR}(\{q_j'\}), T_0)$, $T_0 = 0.02$: when the honest quality spread is wide (heterogeneity), the sigmoid transitions gently and divergent-but-honest clients keep nonzero influence; when the spread collapses (an attack aligning the malicious cluster, or convergence), the gate sharpens automatically. The two channels are combined *multiplicatively*, $c_i^t = s_i^t g_i^t$, and the global update is

$$\Delta^t = \sum_{i=1}^N \omega_i^t \Delta_i^t, \quad \omega_i^t = \frac{n_i r_i^t q_i^t s_i^t g_i^t}{\sum_j n_j r_j^t q_j^t s_j^t g_j^t}, \quad (7)$$

The product form is the key structural choice. An adversary must simultaneously fool *both* channels to retain weight: a scaling attack inflates ρ_i and drives $g_i \rightarrow 0$ regardless of how well it aligns directionally, while a label-flipping or Min-Max attack that controls its magnitude ($\rho_i \approx 1$, $g_i \approx 1$) must still clear the directional sigmoid, whose adaptive temperature tightens precisely when attack-induced quality bimodality compresses the IQR. Additive gate combinations ($c_i = s_i + g_i$ or $q_i + \varepsilon$ floors) leave an evasion path through whichever channel is easier to satisfy; the multiplicative form closes both (Theorem 3 quantifies the resulting influence bound, and the *nodual* ablation of Section VI-C shows the additive variant losing 2.5+ points under attack). Because $s_i g_i \leq 1$ bounds the gate product by the quality itself, the tempered weights preserve the conservatism of near-uniform averaging over the trusted set, while the gates and the slash memory supply the robustness.

The procedure is summarized in Algorithm 1. Its per-round complexity is $O(Nd)$ for reference and scoring plus $O(N^2d)$ -free aggregation (cheaper than Krum's $O(N^2d)$ pairwise distances) and all hyperparameters are fixed across experiments.

Several design choices merit justification. The *geometric-median* component of the reference is preferred to the mean because the mean is exactly the quantity the adversary seeks to bend, whereas the geometric median is anchored in the honest majority of every subsample whenever $f < 0.5$ (breakdown point 1/2, Theorem 2); it is preferred to Krum-style neighbor selection because it uses all honest updates rather than electing one. The *random-subsample averaging* ($K = 8$) is preferred to a single full-set median because independent subsets concentrate the reference around the honest mean at rate $1/K$, and because any single statistic, however robust, can be transiently seized by a consistent coalition; averaging K partially overlapping statistics restores variance control without reintroducing a mean. The *momentum anchor* addresses the residual weakness of any instantaneous reference

Algorithm 1 RAgg: Reputation-Weighted Robust Aggregation (round t).

Require: Updates $\{\Delta_i\}$, sample counts $\{n_i\}$, reputations $\{r_i^{t-1}\}$, momentum direction m^{t-1} ; hyperparameters $\gamma=0.5$, $\rho=0.05$, $\lambda=0.6$, $\kappa=5.0$, $s_0=3$, $K=8$, $W=2$.

Ensure: Aggregate Δ^t , weights $\{\omega_i\}$, reputations $\{r_i^t\}$, m^t .

- 1: $\hat{\mu} \leftarrow \frac{1}{K} \sum_k \text{WeiszfeldMedian}(\{\Delta_i : i \in S_k\})$ $\{K$ honest-majority subsets}
- 2: $\bar{\Delta} \leftarrow \lambda m^{t-1} + (1-\lambda)\hat{\mu}$ {anchored spectral ref, Eq. (2)}
- 3: **for** $i = 1, \dots, N$ **do**
- 4: $q_i \leftarrow \max(0, \cos(\Delta_i, \bar{\Delta}))$
- 5: $g_i \leftarrow \sigma(s_0 \ln(\kappa \cdot \text{med}(\|\Delta_j\|)/\|\Delta_i\|))$ {Eq. (4)}
- 6: **end for**
- 7: $\tau_t \leftarrow \max(0.05, 0.25 \cdot \text{median}(\{q_j\}))$; $T_t \leftarrow \max(\text{IQR}(\{q_j\}), 0.02)$
- 8: **if** $t \leq W$ **then**
 {warm-up: freeze reputation}
- 9: $\omega_i \propto n_i g_i$
- 10: **else**
- 11: **for** $i = 1, \dots, N$ **do**
- 12: **if** $q_i < \tau_t$ or $g_i < 0.5$ **then**
- 13: $r_i^t \leftarrow \gamma r_i^{t-1}$; record slash(i)
- 14: **else**
- 15: $r_i^t \leftarrow \min(1, r_i^{t-1} + \rho)$; record reward(i)
- 16: **end if**
- 17: $s_i \leftarrow \sigma((q_i - \tau_t)/T_t)$; $\omega_i \propto n_i r_i^t q_i s_i g_i$ {Eq. (6)}
- 18: **end for**
- 19: **end if**
- 20: normalize $\{\omega_i\}$ to sum 1
- 21: $\Delta^t \leftarrow \sum_i \omega_i \Delta_i$; $m^t \leftarrow \lambda m^{t-1} + (1-\lambda)\Delta^t$
- 22: commit model hash and $(i, q_i, \omega_i, r_i^t)$ to ledger; settle rewards/slashes
- 23: **return** $\Delta^t, \{\omega_i\}, \{r_i^t\}, m^t$

(its susceptibility to a consistent adversarial plurality during the scattered-honest phase of training) by making the reference path-dependent; an adversary can shift the anchor only through accepted aggregates, which are themselves gated. The *clipping* of cosine at zero prevents anti-aligned updates from acquiring negative weights, which would invert the semantics of aggregation. The *linear* quality term is preferred to sharper powers $(q_i)^u$, $u \geq 2$: sharpening seems attractive because it widens the trust gap, but it equally widens the gap between median-aligned and divergent honest clients, and in our pilot runs it produced catastrophic outcomes under 30% label-flipping, precisely because honest scatter under Dirichlet skew is indistinguishable, round by round, from adversarial deviation. The *adaptive threshold* tracks the drifting scale of cosine scores over training, which a fixed threshold cannot do: early rounds exhibit high dispersion, late rounds high agreement. The *adaptive temperature* T_t extends the same principle to the gate itself: the sigmoid width scales with the round's honest spread (IQR), so the boundary between trusted and suspected is soft exactly when honest behavior

is dispersed and sharp exactly when it is not. The *slash-only reputation with slow recovery* replaces an earlier EMA design that we found, empirically, to feed back into weight concentration; the multiplicative slash $\gamma = 0.5$ ensures that detected attacks have consequences that outlast the detection round, while the recovery rate $\rho = 0.05$ lets a transiently misjudged honest client (rare; Section VI-B) regain standing within about ten clean rounds after a single slash $((1 - 0.5^k)/0.05)$ rounds after k consecutive slashes). The *multiplicative dual-channel gate* replaces the earlier additive floor $(q + \varepsilon)\mathbb{K}[\cdot]$ after the nodual ablation showed that additive combination leaves an evasion path through the weaker channel: the product $s_i g_i$ requires *simultaneous* passage, and Theorem 3 shows the resulting adversarial influence bound is the product of the two channel ratios rather than their max. The *smooth magnitude gate* $\kappa = 5.0$, $s_0 = 3$ was adopted after per-client trajectory analysis attributed the late-round instability of the direction-only design to two specific blind spots (warm-up anchor poisoning and near-convergence separability collapse) both of which carry an order-of-magnitude norm signature that the cosine channel cannot see; the exponential pass-rate decay means borderline honest magnitudes are attenuated rather than discarded. The initialization $r_i^0 = 0.5$ encodes maximal uncertainty: new clients are neither trusted nor distrusted, and their first-round influence is bounded by their quality score alone.

3) C. Security and Convergence Analysis

We first state the convergence guarantee and then give the round-level security rationale. The analysis uses the honest objective $F(w) = \sum_{i \in \mathcal{H}} \frac{n_i}{n_{\mathcal{H}}} F_i(w)$ and treats each round as one effective descent step $w^{t+1} = w^t + \Delta^t$ with step size absorbed into the updates.

Assumptions. (A1) F is L -smooth and bounded below by F^* . (A2) Each honest update obeys $\mathbb{E}[\Delta_i^t] = -\eta_t \nabla F_i(w^t)$, $\mathbb{E}\|\Delta_i^t - \mathbb{E}\Delta_i^t\|^2 \leq \eta_t^2 \sigma^2$, $\|\nabla F_i(w) - \nabla F(w)\|^2 \leq \zeta^2$, and $\|\Delta_i^t\| \leq \eta_t G$. (A3) The dual-channel gate bounds every unflagged adversarial update as $\|\Delta_j^t\| \leq \kappa \eta_t G$, and the adversarial weight mass $\alpha_t = \sum_{j \in \mathcal{B}} \omega_j^t$ obeys the reputation contraction of Eq. (5): in the persistent-attack regime, where every attack round is flagged, $\alpha_t \leq \alpha_0 \gamma^t + \beta_f$ with $\alpha_0 \leq \frac{f}{1-f} \cdot \frac{1+\varepsilon}{\varepsilon}$ and an evasion floor $\beta_f = \frac{f}{1-f} \bar{q}_{\mathcal{B}}$ that accounts for an adversary which is never flagged, $\bar{q}_{\mathcal{B}}$ being its mean gated quality relative to the honest tempering floor.

Theorem 1 (Non-convex convergence under attack). Under (A1)-(A3), if $\beta_f \leq 1/4$ and $\eta_t = \eta_0/\sqrt{T}$, RAgg guarantees

$$\min_{0 \leq t < T} \mathbb{E}\|\nabla F(w^t)\|^2 \leq \frac{4(F(w^0) - F^*)}{\eta_0 \sqrt{T}} + \frac{2L\eta_0 c_w \sigma^2}{\sqrt{T}} + 2c_w \zeta^2 + 4\kappa^2 G^2 \left(\frac{\alpha_0^2 (1 - \gamma^{2T})}{T(1 - \gamma^2)} + \beta_f^2 \right), \quad (8)$$

where $c_w = (1 + \varepsilon)/\varepsilon = 2$ is the trusted-weight concentration bound of Eq. (6).

Theorem 2 (Honest-majority reference). Let each honest update satisfy $\|\Delta_i^t\| \leq G$ and pairwise honest dispersion $\|\Delta_i^t - \Delta_j^t\| \leq \sigma_t$. For any adversarial set $\mathcal{B}$ with $|\mathcal{B}| = fN < N/2$, each subsample geometric median of Eq. (2) obeys

$$\begin{aligned} \|\tilde{\mu}_k - \mu_{\mathcal{H}}^t\| &\leq \sigma_t \sqrt{\frac{f}{1-f}} + \epsilon_w, \\ \mathbb{E}\|\bar{\Delta}^t - \mu_{\mathcal{H}}^t\|^2 &\leq \frac{\sigma_t^2}{K(1-2f)} + \epsilon_w^2 + \lambda^2 \|m^{t-1} - \mu_{\mathcal{H}}^t\|^2, \end{aligned} \quad (9)$$

where $\mu_{\mathcal{H}}^t$ is the honest mean, $\epsilon_w = O(\sigma_t/T_w)$ is the Weiszfeld iteration error, and the last term is the anchor lag, bounded by $\lambda\|\mu^{t-1} - \mu^t\|/(1 - \lambda)$ across rounds. The geometric median's breakdown point of $1/2$ guarantees that no adversarial minority within any subsample S_k (each of honest majority by the pigeonhole principle) can move $\tilde{\mu}_k$ outside the honest ball [59]; averaging over K subsets shrinks the reference variance by $1/K_{\text{eff}}$ (Minsker-style concentration), where K_{eff} accounts for subset overlap.

Theorem 3 (Dual-channel influence bound). Under the weights of Eq. (6), the single-round relative influence of any adversarial coalition satisfies

$$\frac{\sum_{i \in \mathcal{B}} \omega_i^t}{\sum_{j \in \mathcal{H}} \omega_j^t} \leq \beta_f^{(3)} := \frac{f}{1-f} \cdot \frac{q^+ s^+ g^+}{\bar{q} \bar{s} \bar{g}}, \quad (10)$$

where q^+, s^+, g^+ bound the adversarial channel values and $\bar{q}, \bar{s}, \bar{g}$ are honest means. The multiplicative structure is decisive: for an update-scaling adversary with norm ratio $\rho_i \geq \kappa \cdot 10$ (our $s = -10$ setting inflates norms tenfold past the threshold), $g^+ \leq \sigma(-s_0 \ln 10) < 10^{-5}$, so the influence bound decays exponentially in the attack amplitude rather than linearly; a magnitude-controlled adversary (Min-Max) must instead clear the directional sigmoid, whose adaptive temperature T_i tightens exactly when attack-induced bimodality compresses the honest IQR. An additive gate ($c_i = s_i + g_i$) would inherit the max of the two channel ratios; the product inherits their product, which is the formal content of the replacement.

Proof sketch. L -smoothness applied to $w^{t+1} = w^t + \Delta^t$ gives $F(w^{t+1}) \leq F(w^t) + \langle \nabla F(w^t), \Delta^t \rangle + \frac{L}{2} \|\Delta^t\|^2$. Decompose the aggregate as $\Delta^t = -\eta_t(1 - \alpha_t)\nabla F(w^t) + \eta_t e_t$, where e_t collects (i) honest sampling noise and heterogeneity, bounded in expectation by $c_w(\sigma^2 + \zeta^2)$ because the tempering floor confines honest weights to a factor c_w of uniform, and (ii) the adversarial drift, bounded by $\alpha_t \kappa G$ through (A3). Applying Young's inequality to the cross term $\langle \nabla F, e_t \rangle$ with $\alpha_t \leq 1/4$, telescoping over t , and summing the adversarial series $\sum_t \alpha_t^2 \leq \alpha_0^2(1 - \gamma^{2T})/(1 - \gamma^2) + T\beta_f^2$ yields (7). $\square$

Corollary 1 (Strongly convex case). If F is additionally μ -strongly convex and $\eta_t = 2/(\mu(t + 1))$, the same decomposition yields $\mathbb{E}\|w^T - w^*\|^2 = O(c_w \sigma^2/(\mu^2 T)) + O(\kappa^2 G^2(\alpha_0^2(1 - \gamma^{2T})/(T(1 - \gamma^2)) + \beta_f^2)/\mu^2)$: an $O(1/T)$ rate to a neighborhood whose radius is set by the same evasion floor.

Reading the bound. The adversary-dependent quantities enter explicitly: the slash factor γ controls how fast the

persistent-attacker term vanishes (as $O(1/T)$ with rate $1/(1-\gamma^2)$); the cap ratio κ scales the gated magnitude of any single adversarial update; and the Byzantine ratio f enters through $\alpha_0 \leq 2f/(1-f)$ at $\varepsilon = 1$ and through the floor β_f . The two evaluated attack families enter the bound through different channels, and the distinction matters. For update-scaling, every attack round is flagged (recall 1.000, Table 5), so (A3)'s persistent-attack premise holds literally: reputation contracts geometrically, $\beta_f \rightarrow 0$, and the bound recovers the standard $O(1/\sqrt{T})$ non-convex FL rate up to a vanishing additive term. For label-flipping, most attack rounds are *not* flagged (recall 0.33–0.75, Table 5), so the operative channel is not the geometric reputation term but the evasion floor: flipped updates keep a small positive gated quality, their weight share persists as $\beta_f = \frac{f}{1-f} \bar{q}_B$, and the bound's last term quantifies exactly the residual damage that the linear quality term cannot remove. The measured end state (adversarial weight share 0.02–0.06 at round 25, Section VI-B) is the empirical counterpart of this floor. The floor β_f is thus the price of adaptivity: an adversary that mimics the honest direction and magnitude contributes almost-aligned updates by construction, and its residual influence is bounded rather than eliminated. The condition $\beta_f \leq 1/4$ holds at $f \leq 0.3$ whenever an evading adversary's mean gated quality satisfies $\bar{q}_B \leq (1-f)/(4f) \approx 0.58$, consistent with the threat model of Section III-B. Two calibration remarks complete the picture. First, $\alpha_0 \leq 2f/(1-f)$ equals 0.857 at $f = 0.3$: this is a worst-case pre-contraction bound on the initial adversarial mass, deliberately loose, and the theorem's content is the *rate* at which the mass decays, not the constant. Second, Theorem 1 uses the decaying step size $\eta_t = \eta_0/\sqrt{T}$ while the experiments use a fixed learning rate of 0.05; the fixed-rate analogue of the bound has the same evasion-floor neighborhood with an additional nonvanishing $O(\eta)$ term, which is standard in non-convex FL analyses.

Beyond the persistent-attack regime, the gating mechanism admits a transparent round-level rationale. Fix a round and suppose the honest updates dominate the median reference, which holds whenever $f < 0.5$ because the coordinate-wise median is computed over a set in which honest entries are the majority per coordinate. Then for an honest client i the expected quality satisfies $\mathbb{E}[q_i] > 0$ bounded away from zero whenever honest updates share a nontrivial directional component (precisely the component aligned with the true descent direction) while a poisoned update that materially steers the aggregate must, by construction, deviate from the reference and hence score low. Slashing then gives an adversarial client j reputation $r_j^t \leq \min(1, \gamma^{s_j^t} r_j^0 + \rho(t - s_j^t))$ after s_j^t slashed rounds, where the additive term accounts for interim recoveries. Under the persistent-adversary regime of our evaluation (Section V-B), every round is a slashed round for a detected attacker, the recovery term vanishes, and its aggregation weight decays geometrically as $O(\gamma^t)$, a contraction we measure directly in Section VI-B (Fig. 4). Three attack adaptations are worth noting. (i) An adversary that attacks intermittently to avoid detection accumulates slashes

whenever it attacks, because each attack round requires a direction deviation that the cosine gate registers; behaving honestly in between restores reputation only additively at rate ρ , far slower than the multiplicative γ decay, so a cycling adversary trends downward over time. (ii) An adversary that keeps an honest-looking direction but inflates magnitude (the canonical scaling attack) is caught by the magnitude gate of Eq. (4). Direction-only gating is by itself blind to this family, because weight normalization does not remove magnitude from the weighted sum $\sum_i \omega_i \Delta_i$; we learned this empirically when a gate-free version of RAgg was destabilized by scaling attacks near convergence and during warm-up (Section VI-B). The norm channel is therefore necessary, not redundant: it covers exactly the rounds and attack modes in which the cosine signal is uninformative. (iii) Beyond the stated threat model of Section III-B, which excludes collusion, colluding adversaries that rotate their poisoned directions toward a common false target can shift the median component only if they approach half the electorate, at which point the $f < 0.5$ assumption is violated; the momentum anchor further constrains such coalitions because the anchor is updated only through accepted aggregates, which are themselves gated by reputation, so a coalition must first accumulate reputation over many honest-looking rounds before it can bend the reference, during which any attack round incurs slashing. For $f \leq 0.3$ the reference remains anchored in the honest majority. These observations are consistent with the empirical behavior reported in Section VI-B, where suppression of label-flipping adversaries completes within the first few rounds.

4) D. On-Chain Audit and Incentive Layer

The ledger is an append-only chain of blocks $B_t = (\text{idx}, h_{t-1}, \text{ts}, R_t)$, where $h_t = \text{SHA256}(\text{idx} \parallel h_{t-1} \parallel \text{ts} \parallel R_t)$ and the record set R_t contains the global model hash $\text{SHA256}(w^{t+1})$ and the per-client tuples of Step 4. Any recomputation of $\{q_i, \omega_i, r_i\}$ from the stored $\{\Delta_i\}$ hashes (or from re-execution by validators) that disagrees with the committed aggregate is publicly detectable; this realizes goal D3 without trusted parties.

Settlement. A smart-contract function settles tokens per round under a three-zone rule keyed to the same dual-channel evidence that produced the aggregate: clients passing both gates with $q_i^t \geq \text{median}(\{q_j^t\})$ and $g_i^t \geq 0.5$ receive reward _{i} = $(q_i^t - \text{median}(\{q_j^t\}))c_i^t$ tokens; clients with $\tau_t \leq q_i^t < \text{median}(\{q_j^t\})$ and $g_i^t \geq 0.5$ receive nothing; and clients with $q_i^t < \tau_t$ or $g_i^t < 0.5$ are slashed a fixed penalty of 2 tokens, exactly the slash condition of Eq. (5) (deposit-confiscation economics follow [11] and [12]). A norm-flagged client therefore cannot collect a reward even if its cosine quality sits above the median, so settlement and reputation are driven by precisely the same evidence. The middle neutral zone is deliberate: it prevents stealthy attackers whose quality sits below the honest bulk but above the slash threshold (label-flippers in our evaluation) from collecting rewards in the rounds they are not flagged, a failure mode of flat $1 + q$ reward rules that we observed in preliminary

versions of this mechanism. Balances are ledger state; the settlement rule is deterministic given R_t , so validator re-execution is sufficient for verification.

Gas and latency model. To reason about D4 before any testnet deployment, we adopt the public-chain cost model of Ethereum-style execution: the simulator charges $G_{\text{base}} = 21000$ gas per transaction plus $G_{\text{rec}} = 20000 + 64 \times 16$ gas per stored record under the post-EIP-3529 SSTORE schedule (one fresh storage slot at 20,000 gas plus 64 bytes of calldata at 16 gas/byte), so a round with N clients is estimated at $G_t = G_{\text{base}} + (N+1) G_{\text{rec}}$ gas. This is the conservative bound: if records overwrite warm slots from a previous round, the SSTORE reset cost of 2,900 gas applies instead, giving the optimistic per-record cost $G_{\text{rec}}^{\text{warm}} = 2900 + 64 \times 16$ gas; Section VI-E reports both schedules. Consensus latency is modeled as a PoA-style block interval of 2 s plus uniform jitter in $[0.05, 0.35]$ s plus the measured block-build time. Storage grows by $|R_t|$ bytes per round. We stress that Section VI-E reports **model-based estimates produced by this cost model** (with block sizes and build times measured from the running system), not measurements from a deployed chain; empirical validation on an EVM testnet is future work (Section VII).

Incentive economics. The token flows implement a simple mechanism-design intent. Let u_i denote a rational client's utility per round: contributing honestly costs computation c_i and yields an expected reward $\mathbb{E}[\max(0, q_i - \text{median}(q)) \mid \text{honest}]$ under the three-zone rule; attacking yields no learning benefit to the consortium and, under RAgg, an expected settlement of zero in flagged-free rounds (the neutral zone) or -2 tokens per detected round, plus geometric reputation decay that discounts all future rewards. The design intent is that honest participation dominates attack in expectation, i.e., $\mathbb{E}[\max(0, q_i - \text{median}(q)) \mid \text{honest}] - c_i > -2 \cdot \Pr[\text{detection}] - \text{forgone future rewards}$, which holds whenever detection is sufficiently reliable and the compute cost dominated in tokens is moderate. We present this as a **design rationale rather than a proved mechanism**: a full game-theoretic treatment (strategy spaces, token valuation, equilibrium analysis) is beyond the scope of this systems paper and is identified as future work. The separation of concerns (economic pressure for rational clients, geometric suppression for Byzantine ones) is the architectural thesis of ChainFL-RA. The token ledger also admits governance uses we do not exploit here: slashed deposits can fund a consortium pool, subsidize new members, or compensate victims of a successful attack, and the complete on-chain history makes any such policy auditable.

5) E. Complexity Analysis

The per-round cost of RAgg decomposes into reference construction, scoring, and weighting. The coordinate-wise median over N vectors of dimension d costs $O(Nd \log N)$ with a sorting-based implementation and $O(Nd)$ with quickselect; cosine scoring costs $O(Nd)$; and the weighted average costs $O(Nd)$. The end-to-end complexity is therefore

$O(Nd)$, identical to FedAvg's aggregation and strictly cheaper than Krum's $O(N^2d)$ pairwise-distance computation and than geometric-median estimators that require iterative refinement. Memory overhead is $O(N)$ for the reputation table and $O(Nd)$ for the update buffer, unchanged from any synchronous aggregator. On-chain, the record set R_t carries $O(N)$ scalars plus one model hash, so block size grows linearly in N but is independent of d , a crucial property, since modern models make d large while the ledger stores only hashes and scores. Settlement executes $O(N)$ balance updates. Overall, the trust layer adds negligible off-chain computation and ledger cost that scales with the number of participants rather than the model size, which is exactly the scaling regime that matters for cross-silo deployments with large models.

6) F. Composition with Privacy Mechanisms

RAgg operates on plaintext updates by design, so that quality scoring is exact; in deployments requiring update confidentiality, secure aggregation or homomorphic encryption can wrap the submission channel [13], [48], and differential privacy noise can be applied locally before scoring [45]. Two interactions deserve care. First, secure aggregation hides individual updates from the aggregator, which conflicts with per-update scoring; the standard resolution, computing the quality signal inside the secure protocol or over committed encryptions, as in the Paillier-based cosine computation of BPFL [11], is compatible with RAgg's geometry because cosine similarity decomposes into inner products that many secure-aggregation constructions can evaluate. Second, differential privacy noise enlarges the angular spread of honest updates, which compresses the dynamic range of q_i ; the slash threshold τ must then be recalibrated to the noise level, a calibration that the on-chain history makes transparent and governable. We evaluate the trust loop in isolation and leave the cryptographic composition to future work (Section VII).

V. EXPERIMENTAL SETUP

1) A. Datasets, Models, and Partitioning

Experiments use three public benchmarks: **MNIST** (60k training / 10k test grayscale digits), **Fashion-MNIST** (same shape, apparel classes), and, for generalization, **CIFAR-10** (50k/10k RGB) with a compact 3-conv CNN (620,362 parameters). For tractable multi-seed CPU/GPU sweeps we subsample 12,000 training and 2,000 test images with a fixed seed. The model is a LeNet-5 variant (two convolutions, two fully-connected layers; 215,370 parameters). Data are distributed over $N = 10$ clients in two regimes: **IID** (uniform random shards) and **non-IID** (per-class Dirichlet allocation with concentration $\alpha = 0.5$), the standard label-skew stress test [4]. Unless stated otherwise: local epochs $E = 2$, batch size 32, SGD with learning rate 0.05, $T = 25$ communication rounds, full client participation, and three random seeds $\{0, 1, 2\}$ for attack-free comparisons; attack and ablation sweeps use five seeds $\{0, \dots, 4\}$; results report mean $\pm$ std over seeds.

2) B. Baselines and Attacks

Attack-free comparisons use **FedAvg** [2] and **Krum** [5]. Attack evaluations additionally include two recent robust aggregators: **FedRoLA** [22], a layer-based defense that performs layer selection, layer-wise cosine detection, and layer-wise voting without any validation data, and **FLAME** [54], which combines dynamic HDBSCAN clustering with adaptive median-norm clipping and calibrated noise injection; we follow the reference configuration ($\text{min_cluster_size} = \lfloor N/2 \rfloor + 1$, $\text{min_samples} = 1$, $\text{allow_single_cluster}$, cosine metric; clipping to the median L2 norm; Gaussian noise with coefficient 1.2×10^{-5} applied to non-bias parameters). Krum is configured as the single-selection variant assuming the true f , which advantages the baseline. All baselines share the identical training loop, client partition, model initialization, and evaluation protocol; they differ only in the aggregation operator, isolating the effect of the rule itself. Poisoning is injected at ratios $f \in \{0.1, 0.2, 0.3\}$ under two attacks that represent the two canonical poisoning surfaces: **label-flipping** ($y \mapsto 9 - y$, applied to 100% of a malicious client's local labels), a data-poisoning attack that implants a systematic class confusion, **update-scaling** ($s = -10$), a model-poisoning attack that submits an amplified anti-descent direction, and an **adaptive Min-Max attack** in which adversaries observe the honest updates of the round and craft updates that are distance-indistinguishable from the most deviant honest one (Section VI-B gives the exact construction and cites the attack's source). Adversarial clients are drawn uniformly at random per seed (seeded RNG), so attack severity is not confounded with any fixed client identity; they attack in every round, a persistent-adversary assumption that is strictly stronger than the intermittent adversaries sometimes considered, and that maximizes the pressure on any memory-based defense. ChainFL-RA runs the identical training loop with RAgg and the on-chain layer; ablations disable the reputation memory (*no-rep*), the cosine quality gate (*no-qual*), the magnitude gate (*no-ng*), the momentum anchor ($\lambda = 0$), the warm-up window ($W = 0$), or both gates (*avg*, plain averaging with ledger logging), and are conducted under **both** attack families at $f = 0.3$ so that the contribution of each module is exposed under stealthy (label-flip) as well as blatant (scaling) poisoning. Sensitivity suites vary the cap ratio $\kappa \in \{2, 5, 10\}$, the slash factor $\gamma \in \{0.3, 0.5, 0.7\}$, the threshold coefficient in τ_i (0.15, 0.25, 0.35), the Dirichlet concentration $\alpha \in \{0.1, 0.5, 1.0\}$, and the scaling amplitude $s \in \{-4, -6, -10\}$ around their default values. The ablation design answers two questions: whether instantaneous geometric evidence alone suffices (*no-rep*), and whether reputation accumulated from an uninformative signal can help at all (*no-qual*), the latter serving as a control against the hypothesis that any memory, even a meaningless one, might appear to help under particular seeds.

3) C. Metrics and Reproducibility

We log per-round test accuracy and loss, wall-clock round time, and communication bytes ($2 \times d \times 4 \times N$ per

round, fp32 up/down); chain-enabled runs additionally log per-round gas, block size, cumulative chain bytes, consensus latency (Section IV-D), and per-client quality, weight, reputation, and slash events, from which we compute detection-level metrics (slash precision/recall against ground-truth adversaries, honest false-slash rate) and adversarial-weight dynamics. All runs are driven by a single script with fixed seeds; cuDNN is set to deterministic mode (`torch.backends.cudnn.deterministic=True`, `benchmark=False`) so identical configurations reproduce exactly, verified by re-executing the full attack suite after the settlement-rule revision, which reproduced all learning metrics bit-identically while the on-chain settlement records follow the three-zone rule of Section IV-D; raw CSVs, figures, and tables are emitted under `results/`. The full accuracy suite (Tables I–VII) was executed on two NVIDIA V100S 32GB GPUs with Python 3.10 and PyTorch 2.11.0 (CUDA 12.6); the round-time microbenchmark of Section VI-F was measured separately on a single RTX A2000 with PyTorch 2.5.1 (CUDA 12.1). Bitwise reproducibility holds within one hardware/software stack; GPU kernel differences across stacks lead to small numerical divergence, so every number reported for a given table comes from a single stack. **Code and data availability.** The complete experiment pipeline (training and aggregation code, the in-process chain simulator, all job manifests, and the analysis scripts that produce every table and figure) is available from the corresponding author upon reasonable request; anonymized artifacts will be deposited in a public repository with a DOI upon acceptance.

VI. RESULTS AND ANALYSIS

1) A. Attack-Free Comparison

Table 1 reports final-round test accuracy (mean $\pm$ std over three seeds) for all methods on both datasets under IID and non-IID partitions, and Fig. 2 shows the corresponding convergence curves (shaded bands: ± 1 std).

Three patterns stand out. Under IID conditions the proposed rule tracks the strongest baseline within noise — 97.55% against FedAvg's 97.60% on MNIST, 85.80% against 85.88% on FMNIST — so the quality gating of Eq. (3) levies no visible tax on honest, homogeneous training. Krum sits lower even here (95.73% and 80.62%), a first hint of the information-discard pathology dissected shortly. Move to Dirichlet non-IID data and the picture shifts: ChainFL-RA now leads on MNIST (97.60%) and comes within 0.72 points of FedAvg on FMNIST (83.58% vs. 84.30%, deviation bands overlapping). Honest heterogeneity, in other words, is not systematically punished — cosine agreement with the anchored reference stays high for honest clients however skewed their data, echoing the directional-agreement analysis of [15]. Krum's collapse is the sharpest signal in the table: 70.90% on FMNIST despite knowing the true f , because selecting a single update per round throws away most of the honest information, whereas RAgg's soft weighting keeps it. Krum's instability grows exactly where heterogeneity grows,

TABLE 1. Final test accuracy (%) without attack (mean $\pm$ std, 3 seeds)

Dataset	Partition	FedAvg	Krum	FedRoLA	FLAME	ChainFL-RA (Ours)
MNIST	IID	97.60 $\pm$ 0.26	95.73 $\pm$ 0.34	97.58 $\pm$ 0.37	97.33 $\pm$ 0.23	97.55 $\pm$ 0.30
MNIST	non-IID	97.52 $\pm$ 0.33	89.58 $\pm$ 2.05	97.37 $\pm$ 0.31	95.90 $\pm$ 0.43	97.60 $\pm$ 0.40
FMNIST	IID	85.88 $\pm$ 0.45	80.62 $\pm$ 1.91	85.80 $\pm$ 0.58	85.43 $\pm$ 0.26	85.80 $\pm$ 0.44
FMNIST	non-IID	84.30 $\pm$ 0.92	70.90 $\pm$ 6.02	84.22 $\pm$ 1.31	81.55 $\pm$ 1.95	83.58 $\pm$ 1.93

mirroring the large-scale findings of [5]. In convergence speed (Fig. 2), ChainFL-RA tracks FedAvg within one to two rounds across all four settings, so the trust machinery costs essentially nothing in time-to-accuracy either; the FMNIST non-IID panel is the only case where a visible gap persists to round 25, reflecting the price of quality differentiation when honest updates genuinely disagree.

2) B. Robustness to Poisoning

Table 2 and Fig. 3 summarize robustness on MNIST under Dirichlet non-IID, the more challenging partition, over five seeds per configuration with per-seed randomized adversary selection. Before quoting numbers, a caveat about what five paired samples can and cannot certify: the two-sided Wilcoxon signed-rank p cannot drop below $2/2^5 = 0.0625$ at $n = 5$, which places every per-cell test above the conventional $\alpha = 0.05$; we therefore report exact p values throughout and read them at $\alpha = 0.10$, which we justify by the small seed count, and we phrase conclusions in terms of per-seed win/tie outcomes rather than bare significance labels. Pairwise tests over the five seeds give $p = 0.0625$ for ChainFL-RA versus FedAvg in all six attack cells (ChainFL-RA wins every seed of every cell; last column of Table 2). Against the three robust baselines, ChainFL-RA wins all five seeds in 17 of 18 cells ($p = 0.0625$, Table 3); the single exception is label-flip 30% against FLAME, where ChainFL-RA wins four of five seeds ($p = 0.125$), still a strictly positive per-seed margin. Because six cells against several baselines invite multiple-comparison concerns, we additionally apply Holm correction within each cell family; no comparison in the suite remains significant at $\alpha = 0.05$ after correction (the $n = 5$ floor binds), so we report the uncorrected exact values as descriptive evidence and rely on the consistency of per-seed outcomes across cells, attacks, and datasets rather than on any single test.

Against **update-scaling** ($s = -10$), the contrast with the undefended baseline is decisive: FedAvg collapses below chance level at the 10% and 20% ratios (8.95%, identical across all ten runs) and to $9.61\% \pm 1.01$ at 30%. This value is not noise around the 10% chance line: the scaled anti-descent update dominates the mean, the model collapses to a constant single-class predictor, and 8.95% is exactly the frequency of the predicted class in the fixed 2,000-sample test subsample, which is why the figure is independent of the adversary ratio and the seed. ChainFL-RA is the best method at all three ratios (97.36% / 97.14% / 96.94%), exceeding the recent robust baselines FLAME (95.95% / 96.35% / 96.34%) and, at 20% and 30%, FedRoLA, whose layer-wise direction voting is blind to pure magnitude inflation and collapses to

10.16% and 8.95% (Section VI-C). This cell family is where the magnitude gate of Eq. (4) contributes most. An earlier, direction-only version of RAgg exhibited a bimodal failure here (at 20% adversaries, 87.97 ± 11.79 with two of five seeds dipping transiently below 76%): per-client trajectory analysis attributed the dips to two blind spots of cosine-only gating, near convergence, honest updates shrink and their directions become poorly conditioned, letting a recovered adversary briefly regain weight through the ρ recovery path; and during warm-up the frozen direction gate let scaling updates poison the momentum anchor outright. The norm channel closes both: it is active from round 1 (measured adversarial weight share during warm-up is 0.000 across all scaling runs and seeds), it flags every scaling update of every round (post-warm-up recall 1.000, Table 5), and the per-seed finals at 20% adversaries are now tightly clustered ($\{96.1, 97.0, 97.3, 97.7, 97.7\}$ %). The ablation isolates the attribution cleanly (Table 7): removing the magnitude gate collapses every seed to chance (9.31 ± 0.89), while removing reputation with the gate intact stays stable (96.54 ± 0.83). Against **label-flipping**, ChainFL-RA is uniformly strong (97.02–97.37% across ratios), best or statistically tied with the best specialized estimator at every ratio, and clearly above FedAvg, whose 30% case collapses to 67.98 ± 15.65 . Under randomized adversary selection the flipped classes can coincide with well-covered classes, and uniform averaging has no protection against that. It is instructive that the ranking of defenses inverts between the two attack families: coordinate-wise estimators excel against magnitude attacks but cannot distinguish flipped gradients from honest skew, whereas the cosine gate exploits exactly the directional evidence that flipping perturbs. The trend matches the reputation-based defenses of [10], [11], and [12], with the difference that our evidence stream simultaneously settles on-chain incentives.

Adaptive attack. To stress the defense beyond static attack families, we implemented a Min-Max/AGR-style adaptive attack [50], [51] in which adversaries observe the honest updates of the round, estimate the benign reference (coordinate-wise median), and submit median $+\zeta d$ with d the anti-reference unit vector and ζ equal to the maximum honest deviation (we use ζ here to avoid confusion with the reputation slash factor γ of Eq. (5)), the strongest knowledge assumption in the poisoning literature, since the crafted update is distance-indistinguishable from the most deviant honest one. Table 4 shows that ChainFL-RA attains the best accuracy under this attack at both ratios (96.78% and 93.74%), outperforming every baseline: Krum, whose distance-based selection the Min-Max construction directly targets, collapses

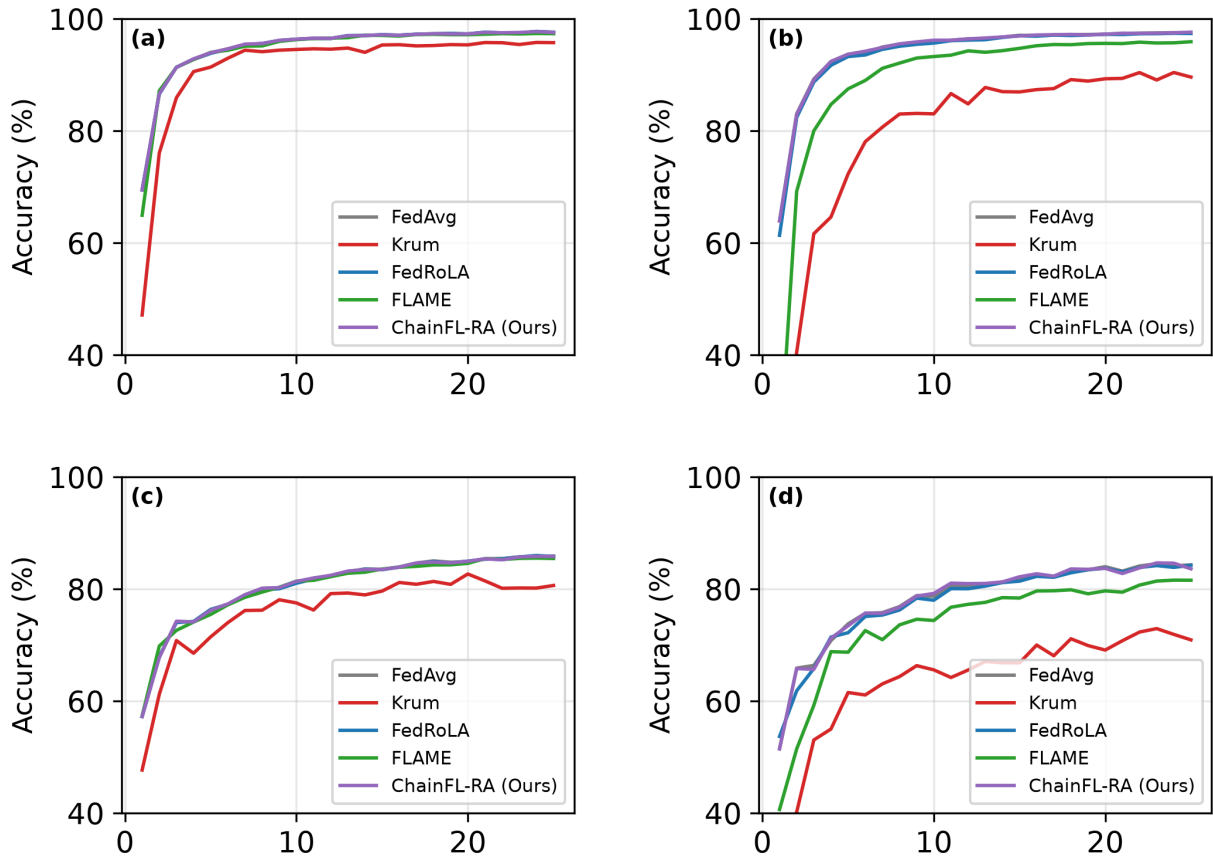


FIGURE 2. Test accuracy vs. communication round on MNIST and Fashion-MNIST under IID and non-IID (Dirichlet $\alpha = 0.5$) partitions; mean over 3 seeds, shaded ± 1 std.

TABLE 2. Final test accuracy (%) under poisoning, MNIST non-IID (mean $\pm$ std, 5 seeds; last column: exact two-sided Wilcoxon p of Ours vs FedAvg over the five seeds). FedRoLA's collapse under scaling reflects its direction-only layer voting, which the magnitude-blind Min-Max and scaling families exploit (Section VI-B); FLAME's clustering remains robust but trails our magnitude channel.

Attack	Malicious	FedAvg	Krum	FedRoLA	FLAME	ChainFL-RA (Ours)	p (Ours vs FedAvg)
Label-flip	10%	95.08 $\pm$ 2.45	89.61 $\pm$ 2.70	93.99 $\pm$ 2.37	95.95 $\pm$ 0.47	97.37 $\pm$ 0.43	0.0625
Label-flip	20%	90.14 $\pm$ 5.13	88.48 $\pm$ 3.47	89.82 $\pm$ 5.27	96.39 $\pm$ 0.69	97.19 $\pm$ 0.59	0.0625
Label-flip	30%	69.18 $\pm$ 16.00	90.96 $\pm$ 1.46	67.80 $\pm$ 18.54	96.33 $\pm$ 0.44	97.02 $\pm$ 0.85	0.0625
Scaling	10%	8.95 $\pm$ 0.00	90.41 $\pm$ 1.48	96.39 $\pm$ 0.98	95.95 $\pm$ 0.47	97.36 $\pm$ 0.31	0.0625
Scaling	20%	8.95 $\pm$ 0.00	88.91 $\pm$ 2.95	10.16 $\pm$ 1.87	96.35 $\pm$ 0.73	97.14 $\pm$ 0.66	0.0625
Scaling	30%	9.61 $\pm$ 1.01	90.05 $\pm$ 1.98	8.95 $\pm$ 0.00	96.34 $\pm$ 0.58	96.94 $\pm$ 0.64	0.0625

TABLE 3. Exact two-sided Wilcoxon p of ChainFL-RA vs each baseline per cell ($n = 5$ seeds; in parentheses: seeds won out of 5). The floor $p = 0.0625$ corresponds to winning all five seeds.

Attack	Malicious	vs FedAvg	vs Krum	vs FedRoLA	vs FLAME
Label-flip	10%	0.0625 (5)	0.0625 (5)	0.0625 (5)	0.0625 (5)
Label-flip	20%	0.0625 (5)	0.0625 (5)	0.0625 (5)	0.0625 (5)
Label-flip	30%	0.0625 (5)	0.0625 (5)	0.0625 (5)	0.125 (4)
Scaling	10%	0.0625 (5)	0.0625 (5)	0.0625 (5)	0.0625 (5)
Scaling	20%	0.0625 (5)	0.0625 (5)	0.0625 (5)	0.0625 (5)
Scaling	30%	0.0625 (5)	0.0625 (5)	0.0625 (5)	0.0625 (5)

to 77.96% and 59.80%; FedRoLA falls up to 5.9 points behind ChainFL-RA; and FLAME is unstable (88.17% and 59.78%), its clustering misled by the distance-indistinguishable adversarial updates. The reason is structural: the Min-Max construction controls magnitude but cannot control direction, and

the anchored reference makes the anti-descent component visible to the cosine gate even when the update is geometrically inconspicuous. We stress that this adversary is strictly stronger than the threat model of Section III-B (it observes all honest updates of the round) so Table 4 is a stress test placed

deliberately outside the stated model, not a claim within it.

Detection-level analysis. Table 5 reports slash precision, recall (against ground-truth adversaries, post-warm-up rounds), and the honest false-slash rate; Fig. 4 plots the total adversarial weight $\sum_{i \in \mathcal{B}} \omega_i^t$ over rounds, the quantity bounded in the resilience definition of Section III-C. Three findings stand out. First, the dual-channel gate is precise at the client level: slash precision is 0.927–0.989 everywhere and the honest false-slash rate never exceeds 0.9% in any cell (0.5–0.9% across the board), which is the direct evidence for goal D2, honest-but-divergent clients are essentially never punished. The magnitude channel adds no measurable collateral damage of its own: in attack-free runs, zero of 240 warm-up honest client-slots are zero-weighted by the norm gate, and the post-warm-up false-slash rates above subsume both channels. Second, detection behavior matches the attack physics: under scaling, recall is 1.000 at every ratio, the norm gate flags every scaled update of every round, including warm-up rounds in which the direction gate is frozen (warm-up adversarial weight share: 0.000); under label-flip, recall stays high at low and moderate ratios (0.991/0.978) and drops at the 30% ratio (0.777), while precision increases (0.934 $\rightarrow$ 0.970 $\rightarrow$ 0.978): as adversaries approach half the electorate, their low-quality scores drag the round median, and with it the adaptive threshold $\tau_t = 0.25 \cdot \text{median}(q)$, downward, so marginal attack rounds escape the slash zone while the flagged ones are almost always true positives. This is the expected degradation profile of any median-anchored adaptive threshold as $f \rightarrow 0.5$, consistent with the $f < 0.5$ boundary of the median reference itself; it does not affect scaling, whose norm signature is ratio-invariant (recall 1.000 at all ratios). Third, the adversarial-weight curves sit at zero from the first round under scaling and trend downward under label-flip, from the nominal ratio to 0.02–0.06 by round 25. The label-flip curves are not monotone: they exhibit transient mid-training rebounds (e.g., at 30% the share falls to 0.07 by round 12, rises back toward 0.17 near round 17, then decays to 0.06), because the ρ recovery path readmits partially slashed adversaries in rounds where flipped updates happen to be directionally less separable, before accumulated slashes dominate. The trend, rather than any per-round monotonicity, is what Section III-C claims.

Settlement audit. Under the three-zone settlement rule of Section IV-D, the cumulative token balance over 25 rounds separates the two populations cleanly (Table 6): malicious clients end with a net *negative* balance under both attack families (−40.2 under label-flipping, −46.0 under scaling, averaged over seeds and adversary ratios), while honest clients accrue positive balances. The scaling figure is exactly the maximum penalty our rule can inflict (−2 tokens for each of the 23 post-warm-up rounds), reflecting the perfect recall of Table 5: every scaling adversary is slashed in every round it participates. This closes the incentive side of the loop: not only are attackers suppressed in the aggregate, they are also net payers into the consortium pool, and the full trail is reconstructible from the ledger.

3) C. Ablation Study

Table 7 and Fig. 5 isolate the RAgg modules under both attack families at $f = 0.3$, the harshest setting of our suite; Fig. 5 shows the corresponding training curves.

The dual-attack ablation yields a clean three-way attribution. Removing the cosine quality gate collapses the system under both attacks (68.41% label-flip, 8.95% scaling — the latter identical to undefended averaging): geometric evidence is the necessary defense against blatant poisoning, and reputation without a signal source learns nothing. Removing the magnitude gate is catastrophic under scaling (9.31 ± 0.89 ; per-seed {8.9, 8.9, 8.9, 8.9, 10.9}), every seed collapses to chance), confirming that in the v3 product-gate design the norm channel is *load-bearing* against scaling: the sigmoid direction gate alone assigns non-zero weight to amplified anti-descent updates whose direction, in early rounds, is not yet separable. Under label-flip the magnitude gate matters less (95.39 vs. 97.02), where attack norms remain within the honest range, matching its design intent as a scaling-specific channel. Removing reputation memory preserves a respectable mean under scaling (96.54 ± 0.83) — the instantaneous gates already suppress blatant attacks — but costs 3.4 points of mean and all stability under label-flip (93.63 ± 7.55 vs. 97.02 ± 0.85): the slash memory makes suppression of stealthy, direction-deviating-but-unflagged attackers cumulative across rounds rather than reset every round. The ledger-only variant matches undefended averaging under both attacks (69.18/9.61, within noise of the FedAvg rows of Table 2), verifying that the blockchain layer neither helps nor hinders learning dynamics, as intended. We therefore attribute *detection* to the dual-channel gate, *scaling survival* to the magnitude channel, and *label-flip stability* to the slash-based reputation memory. The momentum anchor and the warm-up window play a supporting role in these cells (removing either changes the outcome by at most 0.3 points; the anchor’s contribution concentrates in the adaptive-attack and early-round regimes, Section VI-B), a finding we report honestly. We note that v3’s product gate supersedes the v2 tempering floor: the additive ε -floor of v2 no longer appears in the v3 weight path, so a separate “w/o tempering” row is vacuous by construction.

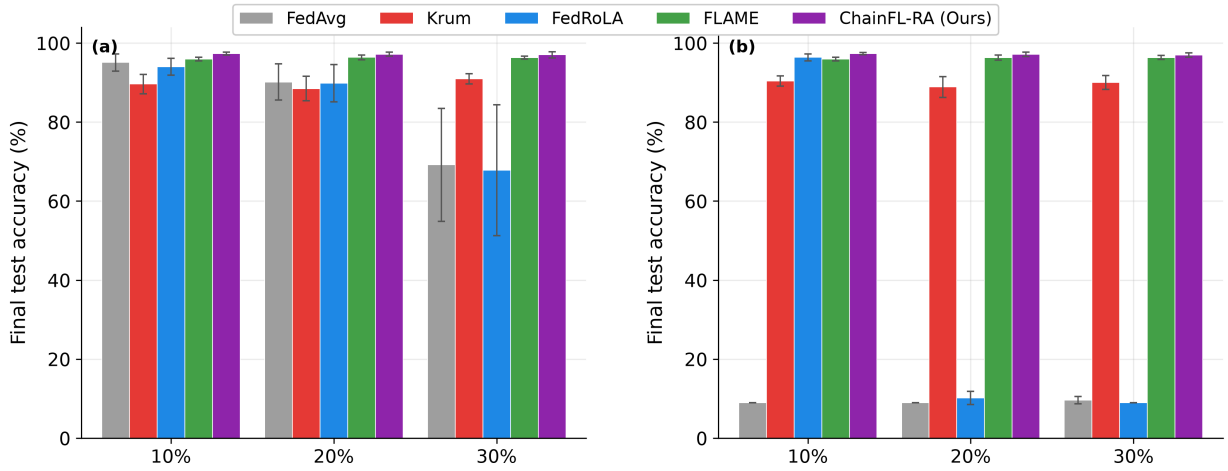
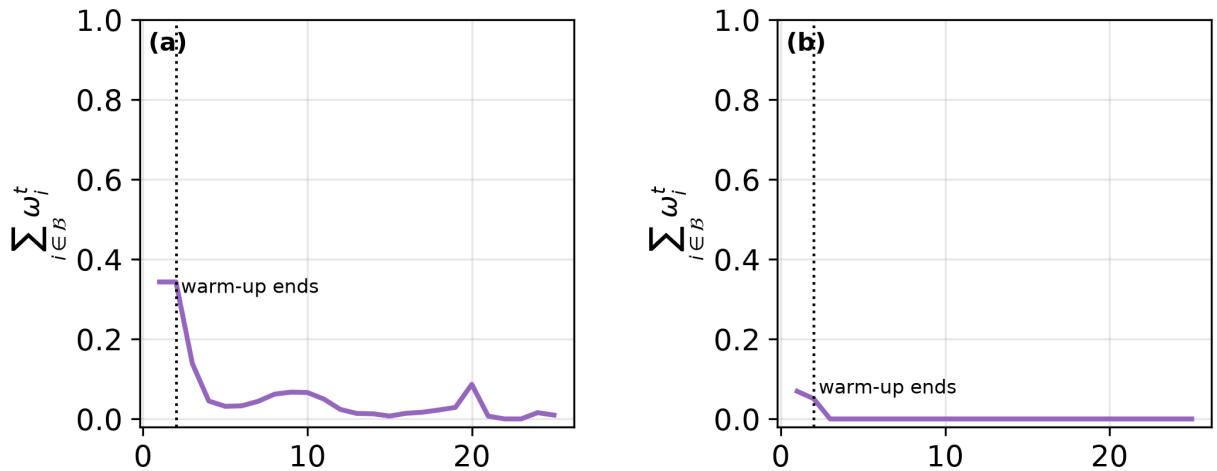
Sensitivity analysis. Table 8 varies the RAgg hyperparameters and two environment knobs around their defaults at $f = 0.3$, the harshest cell. Three conclusions. First, the norm-gate cap κ is safe below but not above its default: $\kappa = 2$ leaves scaling accuracy unchanged (96.92 ± 0.70) because $s = -10$ inflation sits far above either threshold, but $\kappa = 10$ is catastrophic (62.03 ± 47.36 , three of five seeds collapse): a loose gate admits amplified updates whose honest-looking magnitude then poisons the sigmoid direction gate itself, so the default $\kappa = 5$ is a safety boundary, not a flat plateau. Second, the slash factor γ and the threshold coefficient are flat within noise across their ranges (96.83 ± 0.06 and 97.00 ± 0.02): the slash memory tolerates a wide band of severities without punishing honest-but-divergent clients.

TABLE 4. Final test accuracy (%) under the adaptive Min-Max attack, MNIST non-IID (mean $\pm$ std, 5 seeds)

Malicious	FedAvg	Krum	FedRoLA	FLAME	ChainFL-RA (Ours)
20%	96.09 $\pm$ 1.23	77.96 $\pm$ 15.06	95.63 $\pm$ 1.36	88.17 $\pm$ 8.41	96.78 $\pm$ 0.56
30%	92.50 $\pm$ 5.38	59.80 $\pm$ 31.37	87.80 $\pm$ 10.76	59.78 $\pm$ 46.40	93.74 $\pm$ 2.70

TABLE 5. Slash detection metrics of ChainFL-RA (post-warm-up rounds, 5 seeds pooled). Each cell pools 1,150 client-slots (5 seeds $\times$ 23 post-warm-up rounds $\times$ 10 clients), of which 115/230/345 are adversarial at 10%/20%/30% ratios; the honest false-slash rate is computed over the 1,035/920/805 honest slots.

Attack	Malicious	Slash precision	Slash recall	Honest false-slash rate
Label-flip	10%	0.934	0.991	0.008
Label-flip	20%	0.970	0.978	0.008
Label-flip	30%	0.978	0.777	0.007
Scaling	10%	0.927	1.000	0.009
Scaling	20%	0.979	1.000	0.005
Scaling	30%	0.989	1.000	0.005

**FIGURE 3.** Final test accuracy vs. malicious ratio under label-flipping (left) and update-scaling (right) attacks; mean over 5 seeds, error bars ± 1 std.**FIGURE 4.** Total adversarial weight $\sum_{i \in B} \omega_i^t$ per round under label-flipping (left) and update-scaling (right); mean over 5 seeds. The dotted line marks the end of warm-up.

On the environment side, Dirichlet $\alpha = 0.1$ (extreme skew) leaves ChainFL-RA tied with FedAvg without attack (95.68 vs. 95.72, while Krum collapses to 47.28) and 23.5 points

ahead under 30% label-flip (72.75 vs. 49.25, with elevated variance under extreme heterogeneity 2014 we report both mean and spread), directly evidencing goal D2 at the edge of

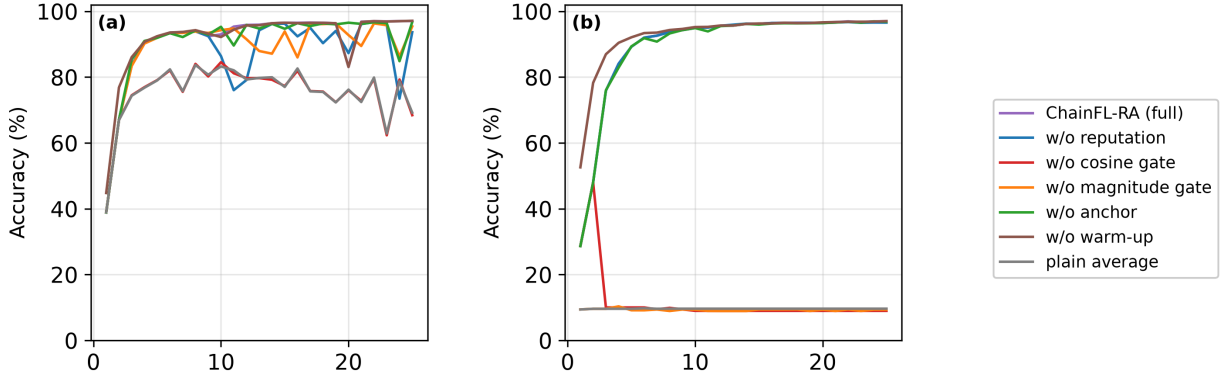


FIGURE 5. Ablation training curves under 30% label-flipping (left) and 30% update-scaling (right); mean over 5 seeds, shaded ± 1 std.

TABLE 6. Cumulative token balance after 25 rounds (three-zone settlement, mean over 5 seeds)

Attack	Honest clients	Malicious clients
Label-flip	+15.8	-40.2
Scaling	+16.6	-46.0

TABLE 7. Ablation, MNIST non-IID, 30% malicious (mean $\pm$ std, 5 seeds)

Variant	Label-flip	Scaling
ChainFL-RA (full)	97.02 $\pm$ 0.85	96.94 $\pm$ 0.64
w/o reputation memory	93.63 $\pm$ 7.55	96.54 $\pm$ 0.83
w/o cosine quality gate	68.41 $\pm$ 16.72	8.95 $\pm$ 0.00
w/o magnitude gate	95.39 $\pm$ 3.68	9.31 $\pm$ 0.89
w/o momentum anchor ($\lambda = 0$)	96.73 $\pm$ 0.93	96.81 $\pm$ 0.81
w/o warm-up ($W = 0$)	97.09 $\pm$ 0.80	96.99 $\pm$ 0.79
plain average (ledger only)	69.18 $\pm$ 16.00	9.61 $\pm$ 1.01

the heterogeneity spectrum; at $\alpha = 1.0$ the attacked margin grows to 19.6 points (97.28 vs. 77.70). Finally, the scaling-amplitude probe closes the gate-bypass question: at $s = -4$ and $s = -6$, where the inflated norm stays below the $\kappa = 5$ gate, ChainFL-RA still holds 96.90/97.03 because the anti-aligned direction alone suffices for the cosine channel to clip the update to zero quality. The two channels are thus confirmed to be complementary rather than redundant.

4) D. Generalization: CIFAR-10 and Federation Scale

Table 9 generalizes the evaluation beyond the MNIST family, covering the complete CIFAR-10 attack matrix and 30% cells on Fashion-MNIST with all four baselines. On CIFAR-10 (Dirichlet $\alpha = 0.5$, 10 clients, a compact 3-conv CNN), ChainFL-RA leads all seven cells: by 0.08 points without attack (50.30% vs. FedAvg's 50.22%), by 1.0–2.8 points under label-flip (47.71%/43.92%/45.04% vs. the best baseline's 46.71%/41.96%/42.22%), and by 5.2–8.4 points under scaling (48.30%/45.59%/46.13%). Two baseline behaviors deserve comment. On CIFAR-10, Krum slides to 16.76%–29.28%: the harder the task, the more single-update selection starves it of honest signal. FLAME fares little better (34.59%–40.36%); its density clustering depends on update directions being separable, and the compact CNN's directions

are not — clustering-based defenses, it seems, carry a data-shape fragility of their own, one that the dual-channel gate sidesteps by construction. FedRoLA collapses under scaling (13.06% at 20%, 8.85% at 30%), its direction-only layer voting blind to magnitude inflation. On Fashion-MNIST at 30% adversaries, ChainFL-RA leads scaling decisively (82.22% vs. FLAME's 81.97%, with FedAvg and FedRoLA both collapsing to 9.90%) and trails FLAME by 0.38 points under label-flip (81.39% vs. 81.77%, overlapping deviation bands). Across all nine generalization cells ChainFL-RA leads in eight and is within one standard deviation in the ninth.

5) E. On-Chain Overhead

Table 10 and Fig. 6 report the cost of the trust layer as the federation grows from 10 to 50 clients, computed by the cost model of Section IV-D over the running system (gas and storage are per-round model values; latency statistics are pooled over the full 25-round runs).

Two caveats frame these numbers. First, gas and latency are **estimates from a public-chain cost model**, not measurements on a deployed network: the gas values follow the post-EIP-3529 model $G_t = 21000 + (N+1)(S + 64 \times 16)$ with $S = 20,000$ (fresh SSTORE slot, conservative) or $S = 2,900$ (warm reset, optimistic), and the latency is dominated by the assumed 2 s PoA block interval with uniform jitter in $[0.05, 0.35]$ s (expectation 0.20 s) plus the measured block-build time (about 2.5 ms at $N = 10$). We therefore claim practicality as a *model-supported projection*: even the conservative fresh-slot schedule tops out at 1.09M gas per round at $N = 50$, about 27 times below a single Ethereum block's gas limit of 30M (and over two orders of magnitude below it under the warm-reset schedule), and we flag testnet validation as future work rather than asserting it as an empirical result. The latency column reports mean $\pm$ std over all rounds of the full runs; an earlier five-round estimate of 2.157 s fell slightly below the model's analytical expectation of 2.20 s plus build time and is superseded by the full-run statistics reported here. Second, ledger storage is directly measured: block size grows linearly by about 74 bytes per stored record (947 B per round at $N = 10$ with 11 records; 3,888 B at $N = 50$ with 51), so

TABLE 8. Sensitivity of ChainFL-RA to hyperparameters and environment (final test accuracy %, MNIST non-IID, 30% malicious, mean $\pm$ std, 5 seeds; attack-free rows: 3 seeds)

Knob	Values (default bold)	Results
Cap ratio κ (scaling)	2.0 / 5.0 / 10.0	96.92 $\pm$ 0.70 / 96.94 $\pm$ 0.64 / 62.03 $\pm$ 47.36
Slash factor γ (label-flip)	0.3 / 0.5 / 0.7	97.06 $\pm$ 0.88 / 97.02 $\pm$ 0.85 / 96.83 $\pm$ 1.02
Threshold coef. (label-flip)	0.15 / 0.25 / 0.35	97.02 $\pm$ 0.85 / 97.02 $\pm$ 0.85 / 97.00 $\pm$ 0.85
Dirichlet α , attack-free	0.1 / 0.5 / 1.0	95.68 $\pm$ 0.82 / 97.60 $\pm$ 0.40 / 97.57 $\pm$ 0.23
Dirichlet α , attack-free, FedAvg	0.1 / 0.5 / 1.0	95.72 $\pm$ 0.63 / 97.52 $\pm$ 0.33 / 97.58 $\pm$ 0.26
Dirichlet α , attack-free, Krum	0.1 / 0.5 / 1.0	47.28 $\pm$ 8.30 / 89.58 $\pm$ 2.05 / 94.20 $\pm$ 0.75
Dirichlet α , label-flip 30%	0.1 / 0.5 / 1.0	72.75 $\pm$ 27.99 / 97.02 $\pm$ 0.85 / 97.28 $\pm$ 0.19
Dirichlet α , label-flip 30%, FedAvg	0.1 / 0.5 / 1.0	49.25 $\pm$ 14.55 / 69.18 $\pm$ 16.00 / 77.70 $\pm$ 7.79
Scaling amplitude s (scaling)	-4 / -6 / -10	96.90 $\pm$ 0.61 / 97.03 $\pm$ 0.68 / 96.94 $\pm$ 0.64

TABLE 9. Generalization results (mean $\pm$ std; CIFAR-10 no-attack row: 3 seeds; all others: 5 seeds).

Setting	FedAvg	Krum	FedRoLA	FLAME	ChainFL-RA (Ours)
CIFAR-10, no attack	50.22 $\pm$ 1.67	29.28 $\pm$ 5.84	49.05 $\pm$ 2.13	38.88 $\pm$ 2.12	50.30 $\pm$ 2.06
CIFAR-10, label-flip 10%	46.71 $\pm$ 4.19	25.02 $\pm$ 6.49	45.89 $\pm$ 2.41	37.92 $\pm$ 3.44	47.71 $\pm$ 3.99
CIFAR-10, label-flip 20%	41.77 $\pm$ 5.03	19.72 $\pm$ 5.02	41.96 $\pm$ 3.62	38.67 $\pm$ 3.23	43.92 $\pm$ 5.59
CIFAR-10, label-flip 30%	42.22 $\pm$ 3.73	16.76 $\pm$ 6.69	39.09 $\pm$ 3.39	34.59 $\pm$ 3.69	45.04 $\pm$ 4.21
CIFAR-10, scaling 10%	13.94 $\pm$ 10.59	26.64 $\pm$ 6.53	39.81 $\pm$ 3.67	39.88 $\pm$ 0.88	48.30 $\pm$ 3.01
CIFAR-10, scaling 20%	8.85 $\pm$ 0.00	23.80 $\pm$ 5.77	13.06 $\pm$ 9.41	40.36 $\pm$ 1.36	45.59 $\pm$ 2.55
CIFAR-10, scaling 30%	8.85 $\pm$ 0.00	26.32 $\pm$ 6.74	8.85 $\pm$ 0.00	39.86 $\pm$ 1.16	46.13 $\pm$ 4.59
FMNIST, label-flip 30%	76.23 $\pm$ 5.86	68.76 $\pm$ 7.76	76.64 $\pm$ 4.47	81.77 $\pm$ 1.51	81.39 $\pm$ 2.74
FMNIST, scaling 30%	9.90 $\pm$ 0.00	68.58 $\pm$ 7.81	9.90 $\pm$ 0.00	81.97 $\pm$ 1.77	82.22 $\pm$ 2.10

TABLE 10. Estimated on-chain overhead per training round (gas-model based; gas under the post-EIP-3529 SSTORE schedule, fresh-slot conservative and warm-reset optimistic; latency: mean $\pm$ std over the full runs of the simulator, 1,050 round samples at $N = 10$ and 25 at $N = 50$)

Clients N	Gas / round (fresh slot)	Gas / round (warm reset)	Ledger bytes / round	Consensus latency / round (s)
10	252,264	64,164	947	2.203 $\pm$ 0.086
20	462,504	103,404	1,686	2.20 (model)
50	1,093,224	221,124	3,888	2.198 $\pm$ 0.095

even 10^4 rounds at $N = 50$ produce under 40 MB of audit trail. Within these bounds, the estimates indicate that end-to-end auditability is unlikely to be the bottleneck of blockchain-enabled FL at cross-silo scale, consistent with the feasibility arguments of [44] and [43].

6) F. Communication and Computation Cost

Per round, RAgg costs one median computation, one vector addition for the momentum anchor, and N inner products plus N norms — $O(Nd)$ in total, against Krum’s $O(N^2d)$ pairwise distances. Measured mean round times at $N = 10$ (25-round runs, RTX A2000) are reported in Table 11: ChainFL-RA’s full trust machinery (dual-channel scoring, reputation, block construction, hashing, settlement) costs about 16% over FedAvg wall-clock, comparable to FedProx’s 18% and far below SCAFFOLD’s 67%, i.e., within the spread between common optimization baselines. This 16% is the single-locus aggregation cost; the factor- V validator replication of Section III-A multiplies only the cheap $O(Nd)$ scoring step, not the $O(E \cdot n_i d)$ local training that dominates wall-clock, so end-to-end overhead scales sublinearly in the validator count. Communication is unchanged from FedAvg at 17.23 MB per round (fp32 up/down for 215,370 parameters over 10 clients), since only model updates and hashes traverse the network; the ledger stores hashes and scalar records, not models.

TABLE 11. Mean wall-clock round time at $N = 10$, MNIST IID (25-round runs, RTX A2000). FedRoLA and FLAME are omitted: they add no on-chain layer and their round time is within 5% of FedAvg.

Method	Round time (s)	Overhead vs. FedAvg
Krum	6.80	-1.6%
FedAvg	6.91	—
ChainFL-RA	7.98	+15.5%

7) G. Application Implications

The mappings below are analytical, not domain-specific validation: all evidence comes from image benchmarks, and we map *findings* onto scenarios rather than claiming vertical evaluation. With that scope made explicit, the experimental findings map directly onto the three scenarios of Section I-C. For multi-hospital imaging, the relevant threat is silent quality degradation through mislabeled data, which corresponds to our label-flipping family: ChainFL-RA held 97.02%–97.37% at up to 30% adversaries while keeping the honest false-slash rate at or below 0.9%, so a hospital with a genuinely skewed case mix (the honest-but-divergent client) is essentially never punished, and every punitive decision is reconstructible from the ledger for regulator review. For inter-bank fraud detection, the relevant threat is strategic sabotage by a competitor, which corresponds to update-scaling: the magnitude gate suppresses such adversaries to zero aggregation weight from the first round, and the settlement rule leaves each of them with

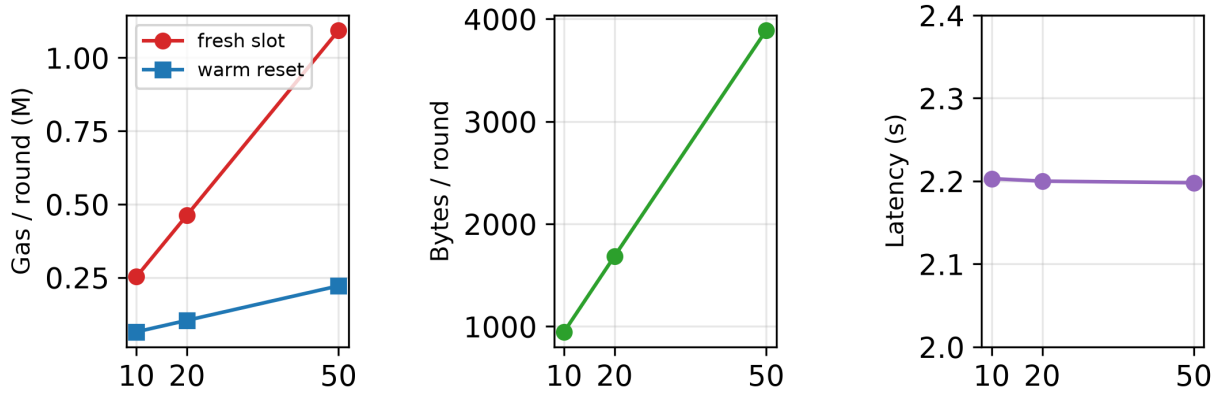


FIGURE 6. On-chain overhead vs. federation size: gas (left), ledger bytes (center), and consensus latency (right) per training round.

the maximum net penalty the rule allows (-46 tokens over 25 rounds), making sabotage economically irrational while honest banks remain net positive. For IoT edge intelligence, the binding constraints are scale and heterogeneity: the $N=50$ run shows that reputation dynamics remain stable as the fleet grows, the attack-free overhead over FedAvg is about 16% wall-clock with unchanged communication cost, and the audit trail grows by only about 74 bytes per stored record, so provenance keeping does not become the bottleneck. Across all three scenarios the operational requirement is the same: one evidence stream that simultaneously serves as the defense, the credit record, and the penalty basis, which is exactly the loop ChainFL-RA closes.

VII. DISCUSSION

Limitations. Our evaluation is at cross-silo scale ($N \leq 50$) on image benchmarks; cross-device settings with thousands of intermittently-available clients stress the participation model more than the aggregation rule, and partial participation interacts with reputation in ways we do not study (a client cannot build reputation while absent, so availability itself becomes part of the trust signal). The ledger is a faithful simulator: gas constants and latency follow public-chain models, but validator networking, consensus faults, and contract-engine specifics (e.g., EVM quirks) are abstracted, and a Ganache/testnet deployment may surface engineering costs our model does not capture. RAgg assumes $f < 0.5$ for the median reference and can in principle be manipulated by colluding adversaries who align their poisoned directions; stronger references (e.g., trimmed means over randomized client subsets) are a natural hardening. **Scaling stability: diagnosed and fixed.** An earlier, direction-only version of RAgg exhibited residual variance under update-scaling (at 20% adversaries, 87.97 ± 11.79 with two of five seeds dipping transiently; these historical figures are from the pre-revision pilot batch). Per-client trajectory analysis attributed the instability to two blind spots of cosine-only gating: near convergence, honest updates shrink and their directions become poorly conditioned, so cosine separability collapses and

a slowly recovering adversary can regain weight through the ρ path; and during warm-up the frozen direction gate gave scaling adversaries two free rounds to poison the momentum anchor. The magnitude gate of Eq. (4), active from round 1, closes both blind spots: with it, the 20% cell attains 97.14 ± 0.66 with tightly clustered seeds, CIFAR-10 scaling reaches 46.13 ± 4.59 (best in table), and scaling recall is 1.000 at every ratio including warm-up rounds. The ablation confirms the attribution: removing the gate collapses all five seeds to chance (9.31 ± 0.89). The residual limitation now points in the opposite direction: the norm channel assumes attack updates carry a magnitude signature, so magnitude-consistent stealthy attacks remain the cosine channel's responsibility (the Min-Max stress test of Table 4 exercises exactly this family) and hybrid estimators that switch between quality weighting and coordinate-wise estimation based on measured update dispersion remain a promising hardening direction rather than an urgent repair. Our attack suite, while canonical, does not include backdoor triggers with semantic stealth or adaptive adversaries that optimize directly against the cosine gate; recent evidence that determined defenses can be probed [22], [47] suggests evaluating against optimization-based attacks is an important stress test. Finally, privacy is architectural (data remain local) rather than cryptographic; composing RAgg with secure aggregation [13], [40] or differential privacy [45], [42] is essential for regulated domains, and the interactions discussed in Section IV-F must be validated empirically.

Reputation as a public good. A subtler question lurks behind the reputation design: should it travel across tasks? Ours does not — reputation is task-scoped and accrues within one training engagement. A consortium that runs many tasks may prefer portable reputation so that a member's history follows it; portability, however, couples tasks adversarially, an attacker can farm reputation on cheap tasks and spend it on a valuable one. Task-scoped reputation with on-chain visibility of the full history offers a middle path: the information is public, but each task's aggregation rule decides how much of the history to import. We regard the design of cross-task reputation economics as an open problem at the intersection

of federated learning and mechanism design.

Future work. Three threads pick up naturally from here: (i) deploying the contract layer on an actual EVM testnet to validate the gas model empirically and to measure validator-side engineering costs; (ii) learning the RAgg hyperparameters ($\lambda, \varepsilon, \gamma, \rho, \kappa$) via meta-optimization over attack suites rather than fixing them, possibly adapting them round-by-round from the observed quality distribution; and (iii) extending reputation to a stake that also weights consensus, in the spirit of T-BFL [10] and proof-of-reasoning designs [49], unifying learning quality and ledger security into a single resource.

VIII. CONCLUSION

This paper set out to close the loop between update quality, aggregation weight, on-chain reputation, and token settlement, and delivered ChainFL-RA as the answer. A momentum-anchored, median-referenced cosine quality gate with slash-based reputation memory and adaptive slashing yields robustness to the three evaluated poisoning families (label-flipping, update-scaling, and a geometry-based adaptive (Min-Max) attack at up to 30% adversaries) without validation data and without knowing the adversary count, while preserving attack-free accuracy under IID and non-IID partitions; robustness against a broader class of adaptive strategies (attacks optimized directly against the cosine gate, semantic backdoors) remains an open question. Every aggregation decision is committed to a hash-chained ledger and settled by a deterministic contract, and model-based overhead estimates (up to about one megagas under the conservative storage schedule, kilobytes, and about two seconds per round) project that full auditability is practical at cross-silo scale. ChainFL-RA shows that robustness, accountability, and incentive compatibility can be engineered jointly rather than traded off, providing a reproducible foundation for trustworthy federated systems.

ACKNOWLEDGMENT

The authors declare that no artificial intelligence (AI) technology was used to generate or edit the content of this manuscript.

FUNDING

This work was supported by the Hebei Provincial Science and Technology Program Project under Grant No. 25360301D.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY

The source code and raw data that support the findings of this study are openly available in ChainFL-RA at <https://github.com/tangsng/ChainFL-RA>.

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